

附:学术作品集（论文3篇）

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注：

1. 第一篇《基于神经网络的消费者购买决策预测模型及其在营销中的应用》（第三作者）已被EI源刊《*International Journal for Housing Science and Its Applications*》录用。
2. 第二篇《方差自适应驱动的RAVEN-UCB：多臂老虎机非稳态场景下的算法实践》（第一作者）在投 *CCF-C期刊*，Under Peer Review.
3. 第三篇《金融市场情绪与收益间的非线性关系：基于大语言模型的动态自适应建模研究》（第一作者）已被2025 *International Conference on Big Data, Artificial Intelligence and Digital Economy* 录用。

A neural network-based prediction model for consumer purchase decisions and its application in marketing

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Abstract: Neural networks have received increasing attention recently, which provide a relatively effective and simple method for dealing with highly complex problems. In this paper, a neural network-based prediction model for consumer purchase decision is constructed. The quantile regression function can reveal the characteristics of the entire conditional distribution of the response variable, and then using the neural network structure, the nonlinear structure in the factors affecting consumer purchasing can be simulated. The article selects the data related to the sales of notebook computers from January to June 2024 as the research object for empirical research, and the results show that, after comparing and analyzing with the linear quantile regression model prediction method, it is clear that the model of this paper investigates and predicts with a higher degree of accuracy, and with a better goodness of fit. The five quartiles of 0.1, 0.3, 0.5, 0.7 and 0.9 were selected for prediction, word of mouth and quality service, which can promote consumer purchase decision. Higher selling prices date lower consumer purchase decisions. Notebook higher memory does not significantly promote consumer purchases and should be in line with the normal needs of consumers.

Keywords: neural network; quantile regression; prediction model; distributional features

1. Introduction

In the daily operation of enterprises, both online and offline have generated a large amount of user consumption behavior data. These data bring new development

opportunities for enterprises but also make enterprises face great challenges, how to identify high-quality users and channels, optimize the marketing cost has become a pain point for enterprises in various fields [1]. For enterprises, traditional marketing channels have been unable to effectively meet the personalized and diversified needs of users, and the development of the Internet and data mining technology has broadened sales channels for companies [2-3]. Therefore, precision marketing with the analysis of consumer purchasing behavior as the core, relying on Internet technology and intelligent algorithms, is increasingly becoming the focus of attention of major enterprises [4]. In 2020, according to the research data of Puhua Industry Research Institute, more than 60% of the enterprises applied big data algorithms to enterprise marketing [5]. The role of consumer purchase behavior data to help enterprises is becoming more and more prominent, and enterprises have gradually begun to pay attention to adding big data to all aspects of marketing [6]. Therefore, how to fully mine and analyze user consumption behavior data, and make accurate predictions on consumer purchase decisions based on the analysis results has become an important issue that enterprises in various industries need to solve.

In order to improve the accurate prediction of user consumption behavior, scholars at home and abroad have done some research on user consumption behavior prediction methods. Valecha, H et al [7] studied the relationship between consumer product purchasing behavior and changes in parameters such as environmental, organizational, individual, and interpersonal factors, and used unique feature engineering to design a real-time evolutionary random forest classifier to predict consumer purchasing behavior decisions. Wang, Y et al [8] conducted purchase prediction across heterogeneous social networks and proposed a game-theoretic approach based on a stable matching model that combines account profiles and historical behaviors to identify personally linked accounts between social networking sites and online shopping, and the results showed that the approach provided more accurate predictions of consumers' purchasing behaviors in relation to goods. Dou, X [9] studied unbalanced real shopping data on e-commerce platforms and used the cat-boost model with symmetric decision tree as the base learner to analyze and

predict whether a consumer will buy a certain product or not, and proved that the model has a higher accuracy rate.

With the development of machine learning methods, many scholars began to apply machine learning methods to the problem of consumer purchase decision prediction. Gallicchio, C [10] proposed that for RNN (Recurrent Neural Networks) the higher the level of the network the easier the model forgets the learned historical data, the application of hierarchical recurrent neural networks can improve the model's ability to memorize and better train the model results. Rojas, J. S et al [11] compared traditional and incremental machine learning algorithms for modeling user consumption behavior of OTT services with the goal of personalization and finally concluded that the best traditional machine learning prediction algorithm is Support Vector Machine (SVM). Ketipov, R et al [12] used machine learning models (e.g., decision trees and random forests) to accurately predict consumer behavior based on personality traits based on the link between personality dimensions and shopping behavior, which can be used to accurately predict user preferences based on their personality traits. Martínez, A et al [13] used a gradient tree augmentation algorithm to predict whether a consumer will make a purchase in the near future using data containing more than 10,000 customers and a total of 200,000 purchases, and the method achieved an AUC of 95%. Wang, W et al [14] proposed an XGBoost (eXtreme Gradient Boosting) based machine learning model for prediction of user buying behavior decisions which outperforms other machine learning models in terms of F1 scores, ROC values, and accuracy. Guo, L et al [15] researched and developed a method called SeqLearn to predict consumer payment behavior by analyzing consumer behavioral sequence data and verified the feasibility and effectiveness of the SeqLearn algorithm in experiments.

Because of the complexity of the consumer purchase decision prediction problem, a single model often produces overfitting phenomenon, so there are also a number of scholars who use a combination of models to mine and predictive analysis of purchase decision behavior data [16]. Xu, J et al [17] proposed an SE-stacking fusion model based on information fusion and integrated learning to predict consumer purchase

behavior decisions in e-commerce, which outperforms traditional single machine learning models and recommender systems. Xiao, S and Tong, W [18] used a combination of text processing and clustering methods to construct consumer-product matrices and predicted consumers' future purchasing behavior decisions for products in a target category through a logistic regression classifier, and achieved 98% prediction accuracy in the Jingdong competitor dataset. XingFen, W et al [19] improved the traditional XGBoost by combining the Logistic regression algorithm with XGBoost algorithm to predict the purchasing behavior decisions of consumers on an e-commerce website, and the results showed that the fusion method outperforms the single model in all evaluations. Tian, Y et al [20] proposed a consumer behavior decision prediction model based on improved deep neural network model (rDNN), and the results showed that the AUC value of the method reached 0.8422, which solved the problem of low accuracy and unsatisfactory effect of traditional consumer behavior prediction model to some extent. In summary, the research on the problem of consumer purchase behavior decision prediction is still in the stage of continuous development, and scholars at home and abroad have carried out in-depth research on consumer purchase decision prediction by way of model construction, from the initial statistical method to the development of a single machine learning method, and then to the combined model method. However, in specific practical problems, the prediction performance of current methods is not yet very satisfactory.

This paper mainly selects the relevant data of laptop computer sales on the e-commerce platform from January to June 2024 as the target index. The data are cleaned and organized to screen out the relevant factors affecting the online sales volume of laptop computers. The acquired data are analyzed with descriptive statistics on the distribution and the relationship with sales, so as to initially understand the distribution characteristics of the various influences on film and television and the connection between them. In order to ensure that the quantile regression can more accurately ensure the accuracy of the results, this paper introduces the neural network model into the quantile regression model, according to the relevant influencing factors,

at the five quantile points of 0.1, 0.3, 0.5, 0.7, 0.90, to construct the neural network based on the consumer's purchase decision prediction model. The data are brought into the decision prediction model to reveal the nonlinear relationship between the variables. And the simulation modeling effect is analyzed by comparing and analyzing with the linear quantile regression method.

2. Quantile regression model

In real life, in order to study the effect of the explanatory variables on the response variable, a linear regression model is usually employed and fitted using least squares estimation or least absolute deviation estimation. Least squares is a commonly used method of modeling mean regression, which describes the effect of the explanatory variables on the conditional mean distribution of the response variable. Least squares is an unbiased, efficient and computationally simple method, and the best linear unbiased estimates can be obtained using this method only when the error term satisfies the assumptions of homoskedasticity and zero mean. When the error terms obey normal distribution, the least squares estimation and the maximum likelihood estimation are consistent. However, in real problems, it is difficult to obtain the error terms under various assumptions [21].

2.1 Quantile regression parameter estimation methods

Suppose that the random variables $Y, \{y_1, y_2, \dots, y_n\}$, $\{y_1, y_2, \dots, y_n\}$ are random samples of the random variable Y , and that the distribution of the Y function is:

$$F(y) = P(Y \leq y) \quad (1)$$

For any $0 < \tau < 1$, define the τ quantile of the random variable Y as:

$$Q_\tau(y) = F^{-1}(\tau) = \inf(y : F(y \geq \tau)) \quad (2)$$

Where the proportion of the region less than $Q_\tau(y)$ is τ , and $\inf(y : F(y \geq \tau))$ is the smallest element in the set of y for which the condition $F(y \geq \tau)$ holds.

The method of solving for the unknown parameter in least squares regression is to minimize the sum of squares of the residuals, unlike the least squares regression method of solving for the unknown parameter in quantile regression is to minimize the sum of the absolute values of the weighted errors. According to Koenker and Bassett, the quantile regression loss function can be expressed as:

$$\rho_{\tau}(u) = (\tau - I_{(u)})u \quad (3)$$

Where $I_{(u)}$ is a schematic function, $I_{(u)} = 1$ when $u \leq 0$, and $I_{(u)} = 0$ when $u > 0$, then the loss function $\rho_{\tau}(u)$ can be written as:

$$\rho_{\tau}(u) = \begin{cases} (\tau - 1)u & u \leq 0 \\ \tau u & u > 0 \end{cases} \quad (4)$$

Clearly, the loss function $\rho_{\tau}(u)$ is composed of multiple linear segmented functions that take values greater than or equal to 0, and at the same time is discontinuous at the position $u = 0$, and therefore not trivial at that point.

Rewrite equation (4) in the following form:

$$\rho_{\tau}(u) = (\tau - 1)uI_{(u \leq 0)} + \tau uI_{(u > 0)} \quad (5)$$

When $u = Y - \hat{Y}$ is taken, it is obtained by substituting into the above equation:

$$\rho_{\tau}(Y - \hat{Y}) = (\tau - 1)(Y - \hat{Y})I_{(Y - \hat{Y} \leq 0)} + \tau(Y - \hat{Y})I_{(Y - \hat{Y} > 0)} \quad (6)$$

In least squares regression, it is minimizing the sum of squares of the residuals, and in quantile regression, it is minimizing the sum of the absolute values of the weighted errors, i.e., finding the estimate that minimizes $E[\rho_{\tau}(Y - \hat{Y})]$:

$$\min E[\rho_{\tau}(Y - \hat{Y})] = \min \left[(\tau - 1) \int_{-\infty}^{\hat{y}} (y - \hat{y}) dF(y) + \tau \int_{\hat{y}}^{+\infty} (y - \hat{y}) dF(y) \right] \quad (7)$$

Taking the first order derivative of $\hat{y}$ in equation (7) and making it zero gives:

$$(1 - \tau) \int_{-\infty}^{\hat{y}} dF(y) - \tau \int_{\hat{y}}^{+\infty} dF(y) = F(\hat{y}) - \tau = 0. \quad (8)$$

The set $F(y)$, as a distribution function of y , is monotonically non-decreasing and right-continuous, so the set $\{\hat{y} : F(\hat{y}) = \tau\}$ is a non-empty set, and then the

estimation parameter that makes Eq. (7) hold exists. When the solution is unique, that is, there is a unique corresponding point on the distribution function, $\hat{y} = F^{-1}(\tau) = Q_\tau(y)$.

Let the random vector Y , given a matrix of explanatory variables X , the regression equation for $Y|X$ at the τ quantile can be expressed as:

$$Q_\tau(Y|X) = X^T \beta_\tau \quad (9)$$

where β_τ denotes the unknown parameters.

Assuming $\{(y_i, x_i), i = 1, 2, \dots, n\}$ is a random sample, where x_i is a p -dimensional column vector, establish the quantile regression equation:

$$Q_\tau(y_i | x_i) = x_i^T \beta_\tau, i = 1, 2, \dots, n \quad (10)$$

The unknown parameter β_τ is estimated:

$$\hat{\beta}_\tau = \arg \min \sum_{i=1}^n \rho_\tau(y_i - x_i^T \beta_\tau) \quad (11)$$

After obtaining the estimate β_τ of the unknown parameter $\hat{\beta}_\tau$, i.e., the quantile regression equation is:

$$\hat{Q}_\tau(y_i | x_i) = x_i^T \hat{\beta}_\tau, i = 1, 2, \dots, n \quad (12)$$

According to the formula for the estimated value of the unknown parameter β_τ , it can be found that the result of the coefficient β_τ of quantile regression depends on the choice of the quantile point τ . The value of the quantile point τ ranges from 0 to 1, so for different quantile points τ , there will be a corresponding fitting function, which in turn will give different parameter estimates. This also means that a cluster of fitted curves can be obtained for the conditional quantile of the response variable, and these fitted curves can reveal the location and shape characteristics of the distribution. Thus, quantile regression models have advantages in analyzing the distributional characteristics of the data.

2.2 Lasso regression method

The Lasso regression method is based on least squares estimation, so it needs to satisfy the assumption of normal distribution and is sensitive to outliers. So in this context, with the continuous development and application of quantile regression and regularization methods, combined with the practical needs of the problem in the quantile regression model to add L_1 penalty term, proposed the Lasso quantile regression model and gives the effective algorithm of the entire solution path, overcoming the shortcomings of the least squares regression method. Lasso quantile regression is similar to the Lasso regression, the combination of Lasso variable selection and quantile regression function, on the one hand, can realize the variable selection and outliers. Similarly, the combination of Lasso variable selection and quantile regression function can realize variable selection on the one hand and parameter estimation on the other hand [22].

Assuming that there are p explanatory variables $X_1, X_2, \dots, X_p$ and the response variable is Y , if $(x_{i1}, x_{i2}, \dots, x_{ip}, y_i)$ is $(X_1, X_2, \dots, X_p, Y)$ for the sample data where $i = 1, 2, \dots, n$ and $X = (X_1, X_2, \dots, X_p)$, $Y = (y_1, y_2, \dots, y_n)^T$, and Lasso quantile regression can be expressed as:

$$\hat{\beta}_{\tau-Lasso} = \arg \min \left(\sum_{i=1}^n \rho_{\tau}(y_i - x_i^T \beta_{\tau}) \right), \text{ s.t. } \sum_{j=1}^p |\beta_{\tau,j}| \leq t \quad (13)$$

The Lasso penalty function form can be expressed as:

$$\hat{\beta}_{\tau-Lasso} = \arg \min \left(\sum_{i=1}^n \rho_{\tau}(y_i - x_i^T \beta_{\tau}) + \lambda \sum_{j=1}^p |\beta_{\tau,j}| \right) \quad (14)$$

where $\lambda \sum_{j=1}^p |\beta_{\tau,j}|$ is the penalization term, and λ is the penalization coefficient ($\lambda > 0$). the strength of the L_1 penalty term. When the value of the penalty coefficient λ is larger, it indicates that the degree of compression is more severe. As λ increases, the L_1 penalty term will cause some of the coefficients to be compressed to 0, thus realizing the effect of variable selection. Therefore, for

practical problems, it is very important to choose the appropriate penalty coefficients λ . AIC, BIC criterion is usually used in least squares estimation, and the mean quantile loss function is obtained by cross-validation under quantile regression making the mean quantile loss function to be the smallest, and the expression of the mean quantile loss function is as follows:

$$\frac{1}{n} \sum_{i=1}^n \rho_{\tau} \left(y_i - x_i' \hat{\beta}_{\tau, \lambda} \right) \quad (15)$$

where n is the sample size and $\hat{\beta}_{\tau, \lambda}$ is the estimated coefficients under the quantile τ penalty coefficient λ .

3. Consumer purchase decision model based on neural network quantile regression

The neural network quantile regression model contains the neural network model and the quantile regression model at the same time with two major advantages: on the one hand, through the simulation of the nonlinear structure in the system, so that the neural network in the simulation of the system to obtain a more accurate simulation results at the same time, and do not need to set a specific form of the function; at the same time, using the advantages of the quantile method, by choosing different locations of the quantile point to obtain the corresponding conditional quantile method, the Thus, the state distribution of the response variable is more comprehensively and finely portrayed, so as to characterize its state distribution in an all-round way [23]. The neural network quantile modeling process is shown in Figure 1.

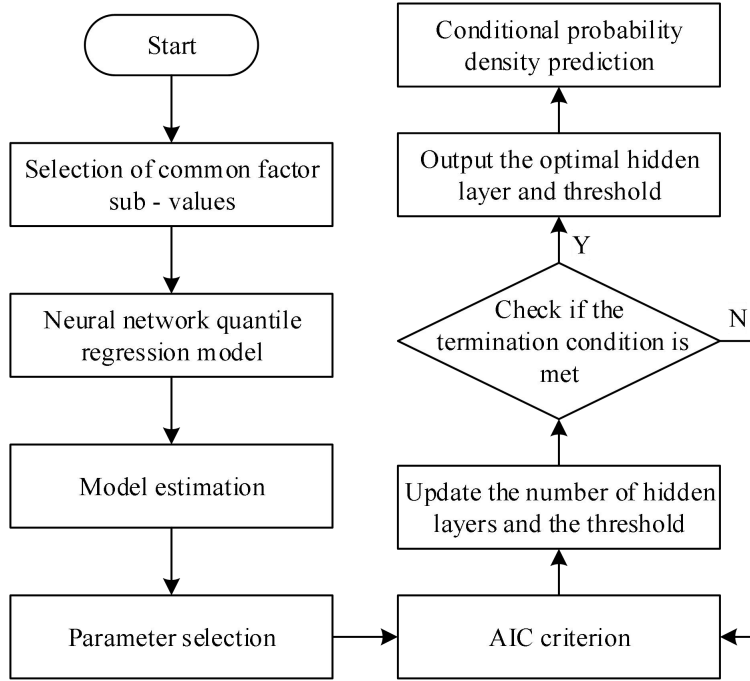


Figure 1 Neural network fractal model process

3.1 Model representation

For the generation of the Quartile Regression by Neural Networks (QRNN) model, it was formed by Taylor's combination of non-parametric regression model properties together in a neural network framework. The model framework underlying the QRNN is set up with a vector of p explanatory variables $x_1, x_2, \dots, x_p$ as the set of explanatory variables, and the response variable is set to be y_t , and the model estimation process is embodied in the explanatory effect of $x_1, x_2, \dots, x_p$ on y_t . Taking the underlying three-layer perceptron neural network analysis, the input layer is the explanatory variables $x_1, x_2, \dots, x_p$, the output layer is y_t , and the node parameter of the implicit layer is set to be J number of nodes, the form of the QRNN model is organized as follows:

$$Q_t(\tau) = f^{(0)}\left(\sum_{j=1}^J \omega_j^{(0)}(\tau) g_j(\tau) + b^{(0)}(\tau)\right) \quad (16)$$

Interpretation of symbols in the model:

τ : the quantile of the response variable y_t .

$f^{(0)}(\cdot)$: the transformation function corresponding to the response variable y_t in the output layer.

$Q_t(\tau)$: conditional quantile of the response variable y_t at different τ .

$\omega_j^{(0)}(\tau)$: measured weights of the layer where the response variable y_t is located.

$b^{(0)}$: similar to the intercept of a function, measures the offset in the response variable y_t estimate data.

$g_j(\tau)$ is an implicit layer node in the network model. Satisfaction:

$$g_j(\tau) = f^{(h)}\left(\sum_{i=1}^p \omega_j^{(h)}(\tau)x_i + b_j^{(h)}(\tau)\right) \quad (17)$$

Where $w_{ij}^{(h)}$ is the implied weight; $b_j^{(h)}(\tau)$ is the implied layer offset, and $f^{(h)}(\cdot)$ is the implied layer transformation function.

In the setting, the number of hidden layer nodes J must not be equal to 1, otherwise the neural network quantile regression model loses its nonlinear characteristics and transforms into a simple linear quantile regression model.

3.2 Model estimation

Combined with the implicit layer node function, this is brought into the total model function. The nonlinear relationship that shrinks between the input and output layers can be expressed by the function as:

$$Q_t(\tau) = f(x_1, x_2, \dots, x_p; \theta(\tau)) \quad (18)$$

A representation of the latter's parameter vector in the above function, which consists of connection weights and offsets:

$$\begin{aligned} &\theta(\tau) \\ &= \left[\omega_{11}^{(h)}(\tau), \dots, \omega_{pJ}^{(h)}(\tau), b_1^{(h)}(\tau), \dots, b_J^{(h)}(\tau), \omega_l^{(0)}(\tau), \dots, \omega_J^{(0)}(\tau), b^{(0)}(\tau) \right]^T \end{aligned} \quad (19)$$

Parameter estimation of the neural network quantile regression model to construct the minimization objective function:

$$\frac{1}{T} \sum_{t=1}^T [\tau - I(y_t < Q_t(\tau))] [y_t - Q_t(\tau)] \quad (20)$$

For the estimation of Eq. (20) in the estimation, in order to prevent the QRNN model from overfitting during the fitting process which leads to the distortion of the model results, it is necessary to introduce a penalty function for overfitting in the objective function. The structure-building optimization function is as follows:

$$\begin{aligned} \hat{\theta}(\tau; \lambda, J) = \arg \min_{\theta} \frac{1}{T} \sum_{t=1}^T [\tau - I(y_t < Q_t(\tau))] [y_t - Q_t(\tau)] \\ + \lambda \frac{l}{pJ} \|\omega^{(h)}(\tau)\|_2 \end{aligned} \quad (21)$$

where $\omega^{(h)}(\tau) \equiv (\omega_{11}^{(h)}(\tau), \omega_{12}^{(h)}(\tau), \dots, \omega_{pJ}^{(h)}(\tau))^T$ is the implicit layer weight vector, $\|\cdot\|_2$ is the paradigm; λ is the penalty parameter.

3.3 Guidelines for parameter selection

In the estimation of the neural network quantile regression model, the optimal values of the penalty function λ value and the number of nodes in the hidden layer J are to be found by referring to the functions described in the previous section to find the optimal point, and the AIC information criterion is introduced to make a judgment based on the need for optimal selection.

The AIC criterion is a model selection criterion that takes into account the goodness of fit of the model and the complexity of the model. Specifically, the AIC criterion is calculated based on maximum likelihood estimation and the number of model parameters [24]. The smaller the AIC value of the model, the better the model's goodness-of-fit and the lower the model complexity. When using the AIC criterion to determine the optimal penalty function value and the number of hidden layer nodes in a neural network quantile regression model, one is essentially looking for an optimal point of trade-off between the model's goodness of fit and model complexity. A larger penalty function value or a smaller number of hidden layer nodes will result in a lower complexity of the model but may reduce the model's goodness-of-fit; conversely, a smaller penalty function value or a larger number of hidden layer nodes may improve the model's goodness-of-fit but the model's complexity will also

increase, and the optimal model can have the smallest AIC value. Define the AIC criterion as follows:

$$\begin{aligned} & AIC(\lambda, J) \\ &= \min \left(\ln \left\{ \frac{1}{T} \sum_{i=1}^T \left[\tau - I(y_i < Q_i(\tau; \theta(\tau, \lambda, J))) \right] \right. \right. \\ & \quad \left. \left. [y_i - Q_i(\tau; \theta(\tau, \lambda, J))] + \frac{p(J+1)}{T} \right\} \right) \end{aligned} \quad (22)$$

In Eq. p is the total number of individuals corresponding to the vector of explanatory variables $x_1, x_2, \dots, x_p$.

3.4 Model predictions

(1) Conditional quantile prediction

Based on the results of the AIC information criterion, the optimal values of the penalty parameter λ and the number of nodes in the hidden layer J in the neural network quantile regression model are screened to construct the estimation of the parameter $\theta(\tau; \lambda, J)$, which will be estimated as $\hat{\theta}(\tau; \lambda, J)$, and substituting $\hat{\theta}(\tau; \lambda, J)$ into Eq. (22), the conditional quantile prediction of the response variable Y_i , $\hat{Q}_i(\tau)$, is obtained [25].

(2) Probability density prediction

On the basis of obtaining the conditional quantile function $\hat{Q}_{Y_i}(\tau | X_i)$, the conditional density in the neural network model into the influence of the change curve, the quantile point τ of the value range of $(0,1)$, the use of asymptotic value and differential extraction in the value range of τ when the continuous value, you can get the condition of the distribution curve F , F to meet the $F(F^{-1}(\tau)) = \tau$, on both sides of the equation, respectively, the calculus of differentiation:

$$\begin{aligned}
\frac{dF(F^{-1}(\tau))}{d\tau} &= 1 \Rightarrow f(F^{-1}(\tau)) \frac{dF^{-1}(\tau)}{d\tau} = 1 \\
&\Rightarrow f(F^{-1}(\tau)) = \frac{d\tau}{dF^{-1}(\tau)} \\
&\Rightarrow f(\hat{Q}(\tau)) = \frac{d\tau}{d\hat{Q}(\tau)}
\end{aligned} \tag{23}$$

where $F^{-1}(\tau) \triangleq \hat{Q}(\tau)$ is the τ quantile.

Conditioning the explanatory variables and discretizing the quantiles yields a simulation of the probability density of the output layer $\hat{f}(\hat{Q}_{Y_i}(\tau))$. Conditioning on X_t and discretizing the quantile τ yields a conditional probability density simulation of the output layer $\hat{f}(\hat{Q}_{Y_i}(\tau | X_t))$.

4. Empirical Analysis of Consumer Purchase Decision Models

4.1 Data sources

This paper takes the influence factors of laptop online sales as an example, in order to make the research have rich empirical significance, the acquisition of data should rely on the e-commerce platform which is widely used by consumers and is representative. Regarding the configuration attributes of electronic products and the basic information of stores, Jingdong Mall has a comprehensive system, and the sales of various products account for a large proportion of the entire market, so it is chosen to crawl the relevant data on Jingdong Mall to provide sufficient data support for the research of this paper.

The dataset is crawled on August 31, 2024, and the crawling range is the relevant data of laptop sales from January to June 2024, and the original data obtained initially is 3838 items. The obtained data is first cleaned and organized to make it usable, because the quality of the data will directly affect the effectiveness of the model.

Regarding the processing of missing values, the data with serious missing features that cannot be filled are directly deleted, and the data with less missing values are filled by using methods such as plurality and mean. Regarding the treatment of outliers, they are directly deleted or treated as missing values, and the data are organized after determining the integrity of the data. After cleaning and organizing,

the final valid data obtained is 3310. The factors affecting the online sales of laptop computers to be selected in this paper are price (x_1), brand (x_2), screen size (x_3), product net weight (x_4), thickness (x_5), core number (x_6), processor (x_7), theoretical endurance time (x_8), warranty life (x_9), commodity rating (x_{10}), logistics rating (x_{11}), and after-sale rating (x_{12}), Commodity rating (x_{13}), logistics rating (x_{14}), after-sale rating (x_{15}), battery capacity (x_{16}), graphics card type (x_{17}), number of battery cells (x_{18}), screen resolution (x_{19}), screen refresh rate (x_{20}), computer type (x_{21}) and memory capacity (x_{22}).

4.2 Descriptive statistical analysis

The actual data on laptop sales span a wide range of values, so to avoid heteroskedasticity, the sales are converted to logarithmic form and the logarithmic sales kernel density is plotted as shown in Figure 2. The figure shows that the logarithmically converted sales show a left-skewed distribution, i.e., the distribution of logarithmic sales is non-normal and concentrated between 2 and 5. In other words, when using sales volume as the dependent variable to build a model, the traditional linear regression model may not have better applicability, and instead, an analysis method that is more in line with the characteristics of the data should be chosen.

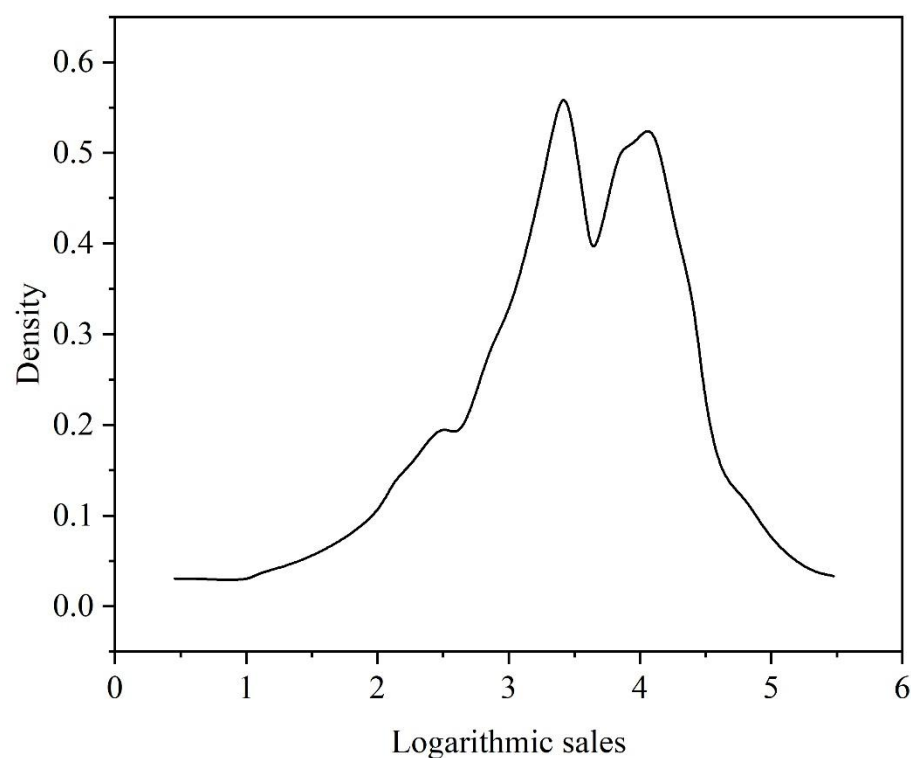


Figure 2 The nuclear density diagram of sales

The configuration level of a laptop computer reflects its excellent performance, which affects the consumer's purchasing decision and thus has an impact on the sales of the item. The laptop attribute variables selected in this paper include categorical and continuous variables, and memory capacity, number of cores, processor and graphics card type are all categorical variables, so the relationship between sales and each factor can be analyzed by drawing box-and-line diagrams, and the results are shown in Figure 3, which shows that there are different degrees of differences in memory capacity, number of cores, processor and graphics card type. The box-and-line plot for memory capacity shows that the mean value of 8GB-16GB memory capacity is relatively large and corresponds to a wide distribution of sales, and the difference in laptop sales between 4GB and 6GB compared to other subgroups is obvious.

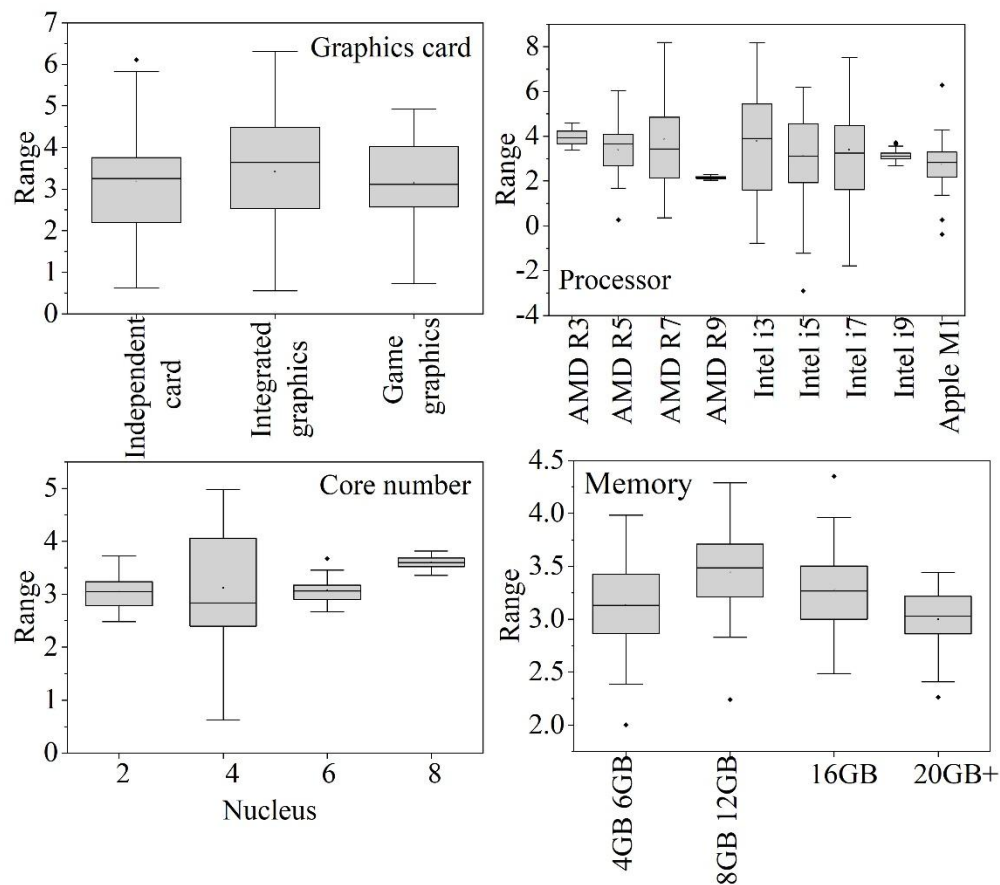


Figure 3 A box diagram of the relationship between factors and sales

In this paper, screen size and battery capacity are transformed into categorical

variables to analyze the impact of these two variables on sales. According to the distribution of the variables, screen size is divided into six categories, namely, below 13.0 inches, 13.0 inches-13.9 inches, 14.0 inches-14.9 inches, 15.0 inches-15.9 inches, 16.0 inches-16.9 inches, and 17.0 inches or more. The battery capacity is divided into three categories, namely, below 30Wh, 30Wh-60Wh, and above 60Wh. The box line plots of the two variables versus sales are plotted separately, and the results are shown in Figure 4, from which it can be seen that either too large or too small a value of the screen size has a significant impact on sales, while the intermediate level has no significant impact on sales. A smaller battery capacity corresponds to fewer sales, indicating that a larger battery capacity can purchase promote sales.

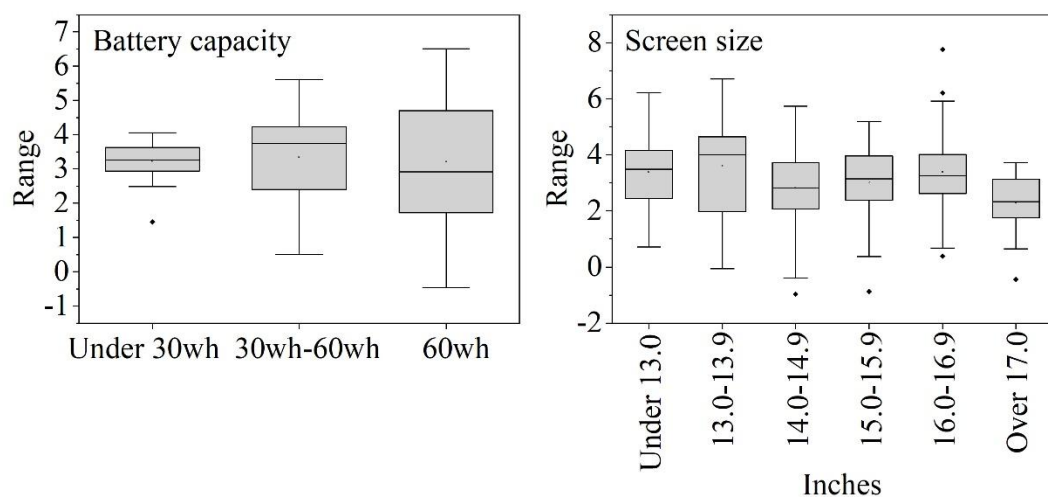


Figure 4 Screen size and battery capacity and sales relationship

The data obtained in this paper covers more than 20 laptop brands, and it is not easy to present all of them, so the brands with larger sample sizes are chosen to plot the kernel density of their prices. The results are shown in Figure 5, the prices of all laptops are roughly between 2000-10000 yuan, which is consistent with the price distribution about laptops in the market. The price distribution graphs of different brands show that the prices of each brand of laptops are mostly skewed. Among them, the prices of Dell, Acer, ASUS, Honor, MI, and HP are mainly distributed around 3,000-6,000 yuan, which indicates that these brands mainly focus on low-end and mid-range computers. And the prices of Apple, Lenovo, and Huawei are mainly distributed around 5,000-10,000 yuan, indicating that these brands sell more high-end laptops.

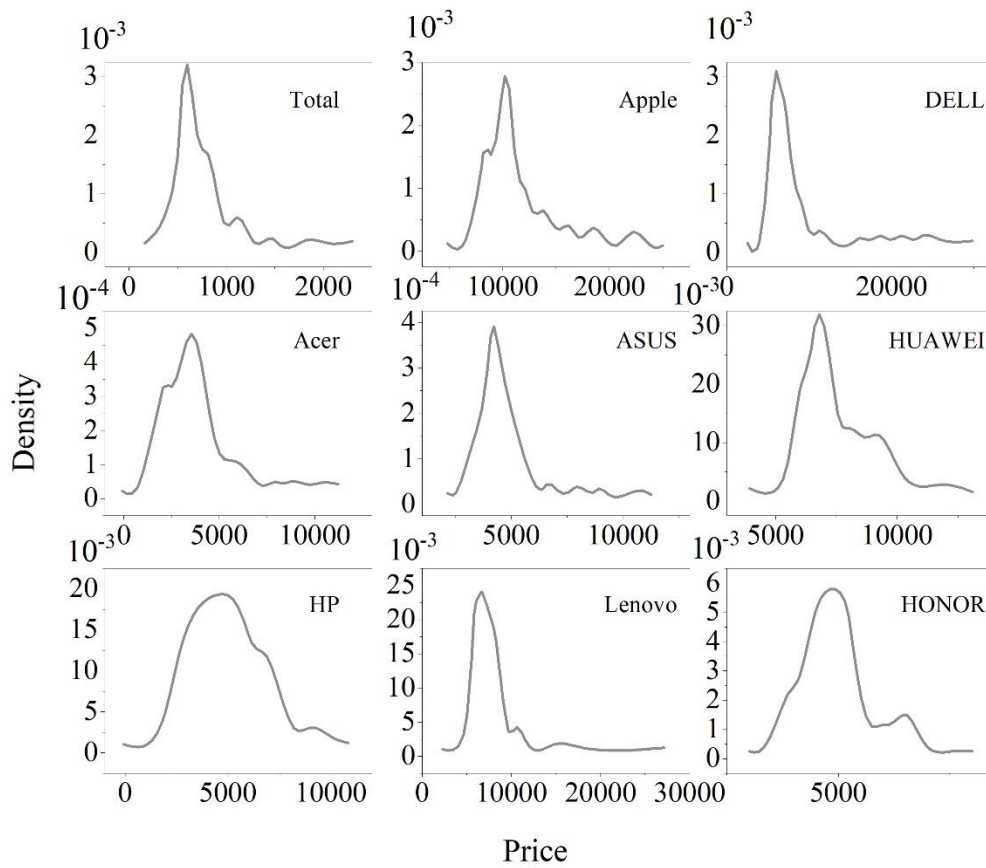


Figure 5 Different brand price distribution

In order to further explore the relationship between price and sales of different brands, the average sales of different brands are compared with the average price, and the comparison results are shown in Figure 6. Where the horizontal coordinate indicates the brand, the vertical coordinate on the left indicates the average sales volume of each brand, and the corresponding image is a bar chart, while the vertical coordinate on the right indicates the average price of each brand, and the corresponding image is a line graph. As can be seen from the graph, with 6000 yuan While Apple, Lenovo, Huawei's products are relatively high-end, their prices are higher and so are their sales.

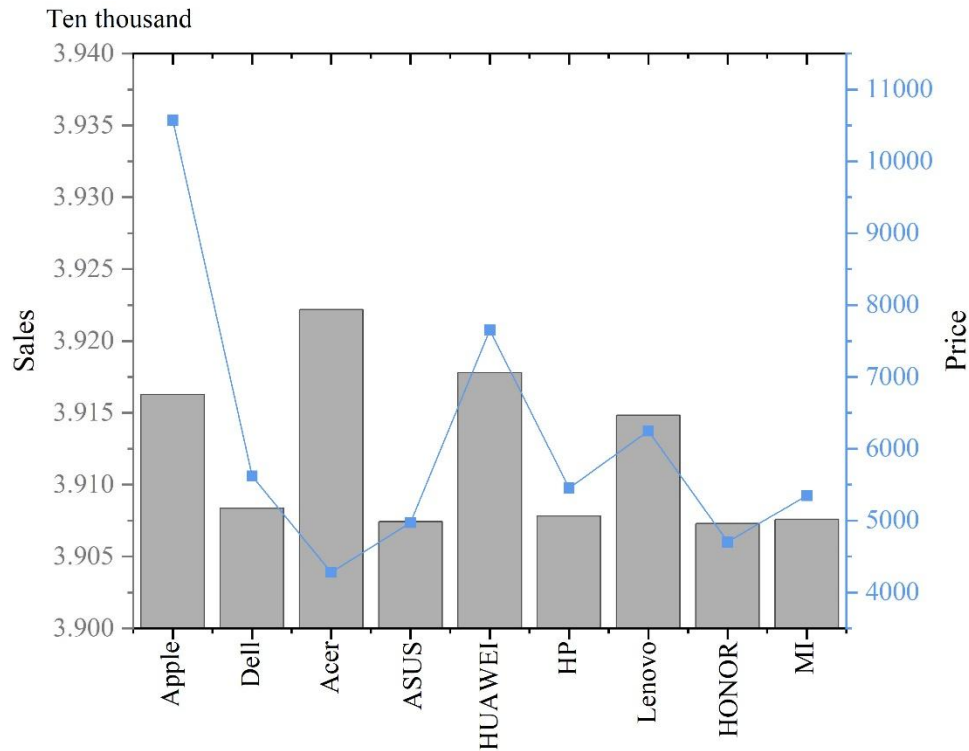


Figure 6 Average and price average diagram of different brand sales

4.3 lasso regression screening variables

In this paper, we use the method of cross-validation and select the ten-fold cross-validation with $K=10$ to determine the parameters of the model. The parameter results are shown in Figure 7, where each curve represents the trajectory of the coefficients of each independent variable, the vertical coordinate is the value of the coefficients, the upper horizontal coordinate is the number of steps, and the lower horizontal coordinate characterizes the degree of sparsity, which can be seen that there is a process of compressing the coefficients of the variables to zero.

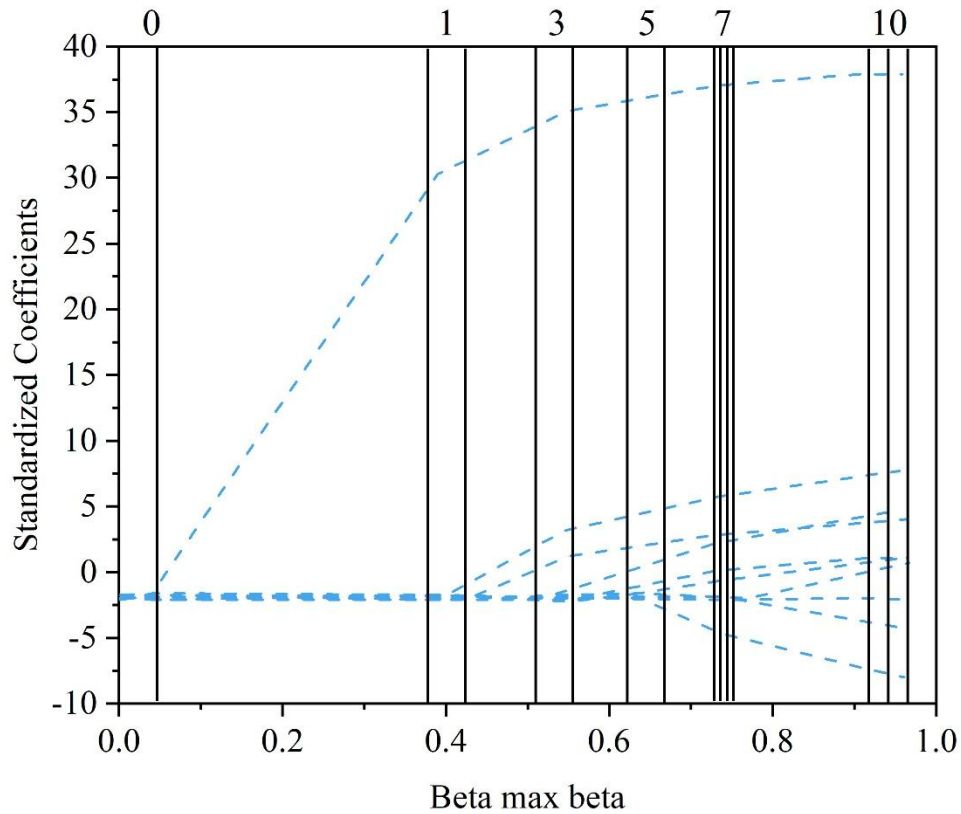


Figure 7 The path diagram of the minimum Angle regression method

4.4 Consumer Decision Modeling Study

4.4.1 Model selection and setup

Secondly, the explanatory variables are analyzed by descriptive statistics, and the results are shown in Table 1, which show that the explanatory variables have different degrees of left or right skewness, and the standard deviation of the price and the number of collections is large, which indicates that the dispersion of the two variables is large. Based on the above characteristics, if the traditional mean regression is still used, the model will not be easy to estimate and the explanatory ability of the model will be worse, as mentioned above, the neural network quantile regression does not exist the assumption of normal distribution and is not sensitive to the perception of outliers in the model, and the constructed model is more robust, so the consumer decision-making model constructed in this paper should choose the quantile regression method. Therefore, the quantile regression method should be chosen for the consumer decision-making model constructed in this paper. In addition to the strong applicability of quantile regression to the model of this paper, it is also very

favorable to the explanation of the consumer purchase decision in this paper. The previous chapter on the theory of consumer purchasing decision established the model of consumer purchasing decision in this paper, in which the explanatory variable is sales volume, sales volume represents the consumer's herd purchasing behavior, which is the consumer's purchasing decision, the higher the sales volume represents the easier for the consumer to make a purchasing decision, and the advantage of quantile regression is that it describes the full picture of the conditional distribution of the explanatory variable, and therefore can describe the different quantile levels of sales volume, and the quantile regression is also very advantageous for explaining the consumer purchasing decision. The advantage of quantile regression is that it describes the full picture of the conditional distribution of the explanatory variables, so that it can describe the different quantile levels of sales, which is very helpful in explaining the consumer's purchase decision.

Table 1 Explain the variable description statistics

Variable name	Mean	Median	Standard deviation	Degree of bias	Kurtosis
Running memory	3.67	4	1.492	0.354	0.911
Screen size	5.51	5.4	0.5683	-1.312	4.025
Core number	3.53	4.1	1.085	-0.952	-0.267
Listing duration	2.45	2	1.112	0.936	0.041
Price	2071.58	1786	1282.5	1.752	6.894
Collection number	2125.84	503	4836.5	5.490	43.135
Evaluation rate	0.3	0.15	0.193	3.024	11.394
Image comment rate	0.22	0.16	0.192	1.52	2.456
Evaluation rate	0.98	1	0.052	-11.482	212.2
Comprehensive score	5.12	5.13	0.08	-1.38	7.714

4.4.2 Estimation of model coefficients

The coefficients of the model were estimated using the `quantreg` package in R language, and five quartiles of 0.1, 0.3, 0.5, 0.7 and 0.9 were selected for the selection of the quartiles, and the mean regression was applied using the same data in the actual

results, and the coefficients estimation results obtained were compared with the estimation results of the quartiles, and the results of the comparison are shown in Table 2.

The neural network quantile regression, although only five quantile points, 0.1, 0.3, 0.5, 0.7, and 0.9, were selected, clearly shows that the impact of each explanatory variable on sales is different at different quantile points, which suggests that there is a heterogeneous effect. Comparing mean regression with neural network quantile regression also reveals that there are differences between the two methods in terms of coefficient significance as well as the direction of influence.

Consumers cannot fully access information about the quality of goods and other information, so word-of-mouth will help consumers reduce the perception of shopping risk and enhance trust, thus promoting consumer purchase decisions. For consumers, the number of favorites has always been the focus of their attention, the more favorites a store has, the more it can reduce the cost of searching for information, thus effectively promoting consumer purchases. For stores with different sales segments, quality service subconsciously affects consumers, thus promoting consumer purchases. Higher memory does not significantly promote the purchase of consumers, memory capacity is not easy to be too large, in line with the normal needs of consumers can be, and too large a memory capacity may bring about a significant increase in price beyond the consumer's psychological expectations and thus unable to promote the purchase. The same goes for core count.

Table 2 Regression table

Variable name		The consumer decision model of the neural network is returned				
		0.1	0.3	0.5	0.7	0.9
Intercept term	-3.025***	-0.725	-0.992	-2.30	-5.298***	-4.82***
memory	-0.056**	-0.040***	-0.032	-0.033	-0.052**	-0.020
Screen size	0.184***	0.093***	0.134*	0.056	0.172***	0.211***
Core number	-0.036	-0.005	-0.007	0.030	-0.035	-0.10***

Listing duration	-0.189***	-0.098***	-0.140***	-0.182***	-0.166***	-0.13***
price	-0.226***	-0.089***	-0.198***	-0.265***	-0.189***	-0.18***
Preferential treatment	0.042***	0.018	0.020	0.056***	0.054***	0.050*
Evaluation rate	1.245***	0.544*	0.613	0.894*	1.45**	1.472*
Evaluation rate	0.189	-0.052	0.046	0.284*	0.23	-0.098
Image comment rate	0.922***	0.325***	0.680***	0.842***	0.87***	0.574***
Collection number	0.475***	0.162***	0.335***	0.462***	0.547***	0.632***
Comprehensive score	0.660***	0.331*	0.400*	0.658***	1.063***	0.951***

4.5. Comparison of forecast results

In the actual modeling process need to divide the data to test the prediction ability, in this paper is mainly selected cross-validation, the data is divided into ten equal parts, in order to select one of them as the test set, the rest of the nine as the training set, for each test set of data to calculate the MAE value, the RMSE value and the AIC value, the total number of calculations ten times, and ten times the results of the averaging, to obtain the final results. the final result.

After dividing the data and applying the neural network method, it is necessary to determine the number of hidden layer nodes, and the number of nodes in the hidden layer is too large, which will lead to the model being too complicated, so the method used in this paper is to select 1 to 5 hidden layer nodes, calculate the AIC values of different hidden layer nodes at different quartiles, and thus to determine the number of nodes in the hidden layer that makes the optimal AIC value, and 3 is selected as the number of nodes in the hidden layer after calculation. After calculation, 3 is selected as the number of hidden layer nodes, and the AIC values of the corresponding neural network quantile regression are calculated on the basis of the number of hidden layer nodes of 3. The comparison results are shown in Table 3.

Through the above table as a whole, the neural network quantile regression is superior to the linear quantile regression in terms of the combined situation of the three indicators. As far as each metric is concerned, for the RMSE metric and the MAE metric, the neural network quantile regression outperforms the linear quantile

regression at all quantile points. Regarding the AIC indicator of model complexity, the AIC values of the two regressions do not differ much at each quantile point, and the neural network quantile regression is superior to the linear quantile regression at the low quantile point 0.1, 0.3, 0.5, and the high quantile point 0.9, while the linear quantile regression model at the middle and high quantile point 0.7 has better goodness-of-fit. Combining the results of these analyses, the goodness of fit of linear quantile regression is comparable to that of neural network quantile regression, but the neural network is more accurate than linear quantile regression in other metrics.

Table 3 Comparison of MAE, RMSE and aic values of the present

Index		RMSE	MSE	AIC
Linear fractional regression	0.1	1.685	1.292	23.956
	0.3	1.204	0.963	22.354
	0.5	0.996	0.785	21.883
	0.7	1.128	0.914	22.189
	0.9	1.782	1.325	21.773
Neural network quantization regression	0.1	1.56	1.22	23.962
	0.3	1.096	0.835	22.48
	0.5	0.944	0.729	22.12
	0.7	1.18	0.84	22.46
	0.9	1.538	1.236	23.79

5. Conclusion

In this paper, the neural network model is combined with the quantile regression model to establish a neural network-based prediction model for consumer purchase decision. Using crawler technology, real data of laptop sales on e-commerce platforms from January to June 2024 are obtained, and the model is empirically tested. Compared with the linear quantile regression model, the neural network quantile regression method has high reliability of estimation results, better prediction ability, and strong nonlinear mapping ability. Moreover, the estimation results of this method have higher precision and better stability. The price factor has a negative and significant effect on consumer purchase. The word-of-mouth factors, such as after-sale rating, product evaluation, logistics evaluation, and after-sale evaluation, show a positive promotion effect. Screen size has a negative and positive influence on consumer purchase respectively.

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From Theory to Practice with RAVEN-UCB: Addressing Non-Stationarity in Multi-Armed Bandits through Variance Adaptation

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Abstract

The Multi-Armed Bandit (MAB) problem faces significant challenges in non-stationary environments where reward distributions dynamically evolve. We propose RAVEN-UCB, a novel bandit algorithm that bridges theoretical rigor with practical efficiency through variance-aware adaptation. Theoretically, RAVEN-UCB achieves tighter regret bounds than UCB1 and UCB-V, with gap-dependent regret $\mathcal{O}\left(\frac{K\sigma_{\max}^2 \log T}{\Delta}\right)$ and gap-independent regret $\mathcal{O}(\sqrt{KT \log T})$. Practically, it integrates three key innovations: (1) Variance-driven exploration via $\sqrt{\hat{\sigma}_k^2/(N_k + 1)}$ in confidence bounds, (2) Adaptive control through $\alpha_t = \alpha_0/\log(t + \epsilon)$, and (3) $\mathcal{O}(1)$ recursive updates for computational efficiency. We validate RAVEN-UCB through a series of experiments across diverse non-stationary patterns—distributional parameter changes, periodic shifts, and temporary fluctuations—using both synthetic environments and a large-scale logistics case study. Results demonstrate consistent superiority over state-of-the-art baselines, confirming the theoretical advantages while highlighting practical robustness in dynamic decision-making scenarios.

Keywords

Reinforcement learning, variance-based methods, Non-Stationary MAB, exploration-exploitation balance, UCB

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1 Introduction

The Multi-Armed Bandit (MAB) problem is a key concept in reinforcement learning and decision theory. An agent sequentially selects among multiple actions, or “arms,” to maximize cumulative rewards. Each arm gives random rewards from an unknown distribution. The goal is to get as many rewards as you can by balancing *exploration* (learning about arms’ reward distributions) and *exploitation* (choosing the arm with the highest estimated reward)[1, 2]. This framework, first formalized by Robbins [3], has led to big changes in areas like online advertising [4], recommendation systems [4, 5], and logistics optimization [6]. Algorithms such as Upper Confidence Bound (UCB), ϵ -greedy, and Thompson Sampling have been studied a lot. They have shown good performance in stationary environments, where reward distributions stay constant over time [7]. However, many real-world situations are different from this assumption. They show *non-stationarity*, where reward distributions $D_k(t)$ evolve dynamically due to things like changing user preferences, market fluctuations, or operational disruptions [8]. This is similar to *concept drift* and shows the need for adaptive MAB algorithms to manage the exploration-exploitation trade-off in dynamic environments. The development of these algorithms is essential for two main reasons. First, they improve performance in current applications. Second, they allow new use cases in environments that are unpredictable. This makes decision-making systems more robust and reliable.

There has been progress in dealing with non-stationarity in MAB problems. But, current approaches have big limits that make them less effective in dynamic settings. We can divide these methods into two main groups. One group uses sliding windows or discounting to deal with mean drift. The other group includes variance information in a static way. Sliding-window methods like Sliding-Window UCB (SW-UCB) [9] use a fixed-size window of recent observations to estimate reward distributions. Discounting techniques like Discounted UCB (D-UCB) [10] use exponential weights to prioritize recent data. These approaches can follow changes in reward means. But they often neglect variance dynamics ($\sigma_k^2(t)$). Variance dynamics are important in environments with changing uncertainty. Recently, algorithms like f-Discounted-Sliding-Window Thompson Sampling (f-dsw TS) [11] have been proposed to address these shortcomings. These algorithms combine sliding-window and discounting strategies with dynamic variance adaptation. This helps improve adaptability in non-stationary settings.

Second, variance-aware algorithms like UCB-V [12] use static variance estimates. They do not capture temporal fluctuations and limiting dynamic exploration. In contrast, RAVEN-UCB employs more flexible parameters with a time-decaying exploration factor. It selects better arms less often, so it is more effective in dynamic environments. Furthermore, sliding-window and variance-aware methods have computational inefficiencies. Maintaining windows or recalculating statistics from scratch incurs $O(n)$ time complexity per update [13], which

is impractical for large-scale, real-time applications like online advertising or logistics. To address these gaps, we propose RAVEN-UCB, a new variance-adaptive bandit algorithm for non-stationary environments. RAVEN-UCB introduces three key contributions:

1. **Variance as an Exploration Signal:** Unlike traditional methods, RAVEN-UCB dynamically adjusts exploration based on real-time sample variance, incorporating a term proportional to $\sqrt{\frac{\hat{\sigma}_k^2}{N_k+1}}$ in its upper confidence bound, where $\hat{\sigma}_k^2$ is the estimated variance of arm k and N_k is the number of times it has been selected (Sec. 3.4). This enables increased exploration during periods of high variance, reflecting greater uncertainty or potential distribution shifts.
2. **Flexible and Robust Parameterization:** RAVEN-UCB uses a highly adaptable parameterization with a time-changing exploration coefficient $\alpha_t = \alpha_0 / \log(t + \epsilon)$ and adjustable parameters β_0 and ϵ . This design ensures responsiveness to environmental changes while maintaining low sensitivity to parameter choices (Sec. 4.2). It ensures stable performance in different non-stationary situations.
3. **Efficient Recursive Updates:** To enhance scalability, RAVEN-UCB uses recursive formulas for updating sample mean and variance, achieving $O(1)$ time complexity per step (Eq. 4–5). This overcomes the computational bottlenecks of naive method, making it suitable for large-scale applications.

We organize the remainder of this paper as follows: Section 2 provides a comprehensive background on the MAB problem, formalizing non-stationary environments through classes such as Distributional Parameter Changes (DPC), Periodic Changes (PC), and Temporary Fluctuations (TF), and reviews existing adaptive bandit algorithms. Section 3 presents the RAVEN-UCB algorithm. We detail its design principles, variance-adaptive exploration, logarithmic decay mechanism, and recursive update formulas, along with theoretical regret bound analysis. Section 4 evaluates RAVEN-UCB’s empirical performance through three experiments: a regret comparison against UCB1, showing an average regret reduction of 84% compared to UCB1; a sensitivity analysis of hyperparameters across different scenarios; and a simulated logistics optimization case study with 100 warehouses, where RAVEN-UCB achieves 68% lower regret than standard UCB. Section 5 concludes by summarizing the main contributions and outlining future research directions, including extensions to contextual bandits and integration with large language models for enhanced decision-making. Detailed derivations and proofs are provided in the Appendix A.

2 Background

2.1 Multi-Armed Bandit Problem

Since it was first introduced in the 1950s by Robbins [3], the Multi-Armed Bandit Problem has been established as a fundamental framework for sequential decision-making under uncertainty. In a typical MAB setting, an agent selects one of K arms at each time step t , receiving a reward $r_k(t) \sim D_k(t)$. And $D_k(t)$ is the reward distribution for arm k [11]. Different policies have been developed to determine which arm to select at each time step in the multi-armed bandit problem. The most studied in the scientific researches are:

- **Upper Confidence Bound (UCB):** This Algorithm picks the arm with the highest sum of the estimated mean reward and an uncertainty term, scaled by a parameter α to control exploration versus exploitation. The chosen arm a is determined as $a \leftarrow \operatorname{argmax}_{k \in \mathcal{K}} \{\hat{\mu}_k(t) + \alpha \cdot f(\hat{\sigma}_k(t))\}$, where $\hat{\mu}_k(t)$ is the estimated mean reward for arm k at time t , $\hat{\sigma}_k(t)$ is the estimated standard deviation, and f scales the standard deviation.[14]
- **ε -greedy:** This strategy explores by randomly selecting an arm with probability ε , while exploiting the arm with the highest estimated mean reward with probability $1 - \varepsilon$. [15, 16]
- **Thompson Sampling:** A Bayesian method that maintains a posterior distribution for each arm's mean reward, sampling from these distributions at each step and selecting the arm with the highest sampled value.[17]

The methods above have been extensively analyzed, and lots of theoretical results guarantee their convergence (i.e., regret bound) to the optimal solution in stationary environments. [12, 18] A stationary setting is defined as an environment in which the reward distribution D_k for each arm k is assumed to be stationary does not change through all the time-steps in T . [11]

MAB algorithms are used in many areas. Important application areas include online advertising, recommendation systems, and logistics and supply chain optimization. In online advertising, MAB algorithms help to optimize ad selection in real time to maximize click-through rates and conversion rates. For instance, Jahanbakhsh et al.[19] modeled ad selection as a MAB problem, dynamically identifying high-performing ads using online learning techniques. Similarly, Nguyen-Thanh et al.[20] proposed a UCB-based recommendation strategy that addresses large action spaces and non-stationary user preferences, significantly improving user engagement. In recommendation systems, MAB algorithms address challenges such as the cold-start problem and dynamically evolving user preferences[21]. Ding

et al.[22] applied ϵ -greedy, Thompson Sampling, and UCB algorithms to personalize product recommendations on an e-commerce platform, achieving notable improvements in user interaction metrics. Meanwhile, in logistics and supply chain management, MAB algorithms are applied to dynamic pricing, inventory control, and resource allocation. Gao and Zhang[23] developed a UCB-based learning framework to better customer selection in multi-product inventory systems, achieving efficient stock management. In emergency logistics, geometric greedy algorithms were proposed to optimize hub locations and resources distribution under uncertainty, improving how quickly and strongly supply chains respond [24].

2.2 Non-Stationarity in MAB

The MAB problem is a basic idea for making decisions when things are uncertain, and lots of research looks at how it can be used in different situations. Real-world uses often differ from basic ideas. For instance, Wang et al.[25] extend the MAB framework to scenarios where the optimal arm depends on a hidden Markov model (HMM), introducing reward correlations governed by an unknown Markov process. Wang, Wang, and Huang address the challenge of combined and anonymous feedback, where rewards are delayed and mixed across actions, like online advertising [26]. They introduced adaptive algorithms (ARS-UCB and ARS-EXP3) to get the best regret bounds without prior knowledge of delay structures, making MAB stronger in both predictable and challenging settings. These improvements show how MAB can adjust to tricky environments, dealing with things like reward links and feedback waits. In real-world situations, such as logistics, reward distributions often change over time due to factors like traffic conditions, inventory levels, or user preferences, leading to non-stationary MAB problems[11]. This non-stationarity is similar to *concept drift* in machine learning where the conditional distribution $P(y|X)$ changes[27]. This challenges traditional algorithms that rely on fixed distributions. Non-stationary MAB environments are divided based on how reward distributions $D_k(t)$, defined by mean $\mu_k(t)$ and variance $\sigma_k^2(t)$, evolve over time. These are further broken down into three main types:

- **Distributional Parameter Changes(DPC):** The mean or variance changes over time:

$$D_k(t) = D(\mu_k(t), \sigma_k^2(t)),$$

where $\mu_k(t) = \mu_k(0) + \delta_k(t)$, $\sigma_k^2(t) = \sigma_k^2(0) + g_k(t)$, and $g_k(t) \geq 0$.

- **Periodic Changes(PC):** The distribution repeats with period P :

$$D_k(t) = D_k(t \bmod P).$$

- **Temporary Fluctuations(TF):** The distribution deviates briefly and reverts:

$$D_k(t) = \begin{cases} D_k^{\text{normal}}, & t < t_b \text{ or } t \geq t_b + \Delta t, \\ D_k^{\text{blip}}, & t_b \leq t < t_b + \Delta t, \end{cases}$$

Meanwhile, the following table 1 summarizes the specific real-world manifestations, mathematical abstractions, and examples of several Non-stationary MAB environments.

Table 1: Some Non-Stationary MAB Scenarios

Scenario	Cat.	Math. Definition	Real-World Example
Incremental Drift	DPC	$\mu_k(t) = \mu_k(0) + \delta_k t, \delta_k \ll 1$	User preference evolves slowly
Variance Drift	DPC	$\sigma_k^2(t) = \sigma_k^2(0) + g_k(t), g_k(t) \geq 0$	Traffic variability affects delays
Gradual Drift	DPC	$D_k(t) = \begin{cases} D_k^{\text{old}}, & t < t_0, \\ D_k^{\text{old}} \text{ w.p. } 1 - \rho(t); \\ D_k^{\text{new}} \text{ w.p. } \rho(t), & t_0 \leq t < t_1, \\ D_k^{\text{new}}, & t \geq t_1 \end{cases}$	Gradual user migration from old to new product version.
Localized Jump Drift	DPC	$D_k(t) = \begin{cases} D_k(t-1), & k \notin \mathcal{S}(t), \\ \mathcal{U}(\mu_{\min}, \mu_{\max}), & k \in \mathcal{S}(t) \\ \mathcal{U}(\sigma_{\min}^2, \sigma_{\max}^2), & k \in \mathcal{S}(t) \end{cases}$	Edge device resets after network reconnection or resource change.
Periodic Drift	PC	$D_k(t+P) = D_k(t)$	Seasonal demand cycles
Blips / Outliers	TF	$D_k(t) = \begin{cases} D_k^{\text{normal}}, & t \notin [t_b, t_b + \Delta t), \\ D_k^{\text{blip}}, & t \in [t_b, t_b + \Delta t) \end{cases}$	Short-term strike / outage
Add/Remove Arm	TF	$\mathcal{K}(t) = \begin{cases} \mathcal{K}_0, & t < t_a, \\ \mathcal{K}_0 \cup \{k'\}, & t_a \leq t < t_r, \\ \mathcal{K}_0 \cup \{k'\} \setminus \{k''\}, & t \geq t_r \end{cases}$	New warehouse opens / closes

For example, in logistics, parameter changes might show efficiency variations due to traffic. Periodic changes may link to seasonal demand. Temporary fluctuations can mean short-term disruptions like strikes [28]. These types show why adaptive strategies are important for handling non-stationary MAB problems. Recent progress has been exploring large language models (LLMs), such as GPT-3, said Brown et al.[29], to offer new ideas on decision-making in dynamic environments. Discounted UCB (D-UCB) and Sliding-Window UCB (SW-UCB) reduce the impact of old rewards to handle drift[9]. Similarly, there are discounted or windowed types of Thompson Sampling for sub-Gaussian rewards. Meanwhile, the f-Discounted-Sliding-Window Thompson Sampling (f-dsw TS) algorithm uses discount factors and sliding windows to enhance adaptability to concept drift [11]. This approach builds on earlier adaptive strategies. It offers a strong solution for dynamic reward distributions in complex real-world situations. The classic epsilon-decreasing method exists, and SoftMax algorithm, by Velonis and Vergos[30], uses a probability distribution to adjust the likelihood of selecting each arm. It is based on its estimated reward, while sliding-window methods focus on re-

cent data [9]. Again, variance-aware methods like UCB-V guide exploration using variance estimates[12].The limits of current methods and the importance of variance in understanding uncertainty push us to propose RAVEN-UCB. This is a variance-adaptive algorithm. It estimates mean and variance and adjusts exploration over time. This approach improves adaptability in changing environments, especially when distribution parameters change.

3 Methodology

In this section, we show why and how the **RAVEN-UCB** algorithm works. It builds on the classical Upper Confidence Bound (UCB) method and uses variance-based exploration to improve the approach.

3.1 Exploration Signal Based on Variance

Variance helps measure uncertainty in random variable distributions. [31] When sampling is low, the sample mean and variance can change a lot. These changes show where the agent can explore to get better reward estimates. So, the agent uses changes in variance as a signal to explore.

For a given arm k , the sample mean $\hat{\mu}_k$ is calculated as:

$$\hat{\mu}_k = \frac{1}{n_k} \sum_{i=1}^{n_k} X_{k,i} \quad (1)$$

where $X_{k,i}$ represents the reward from the i -th pull of arm k , and n_k is the number of times arm k has been pulled.[32] The sample variance $\hat{\sigma}_k^2$ is given by:

$$\hat{\sigma}_k^2 = \frac{1}{n_k - 1} \sum_{i=1}^{n_k} (X_{k,i} - \hat{\mu}_k)^2 \quad (2)$$

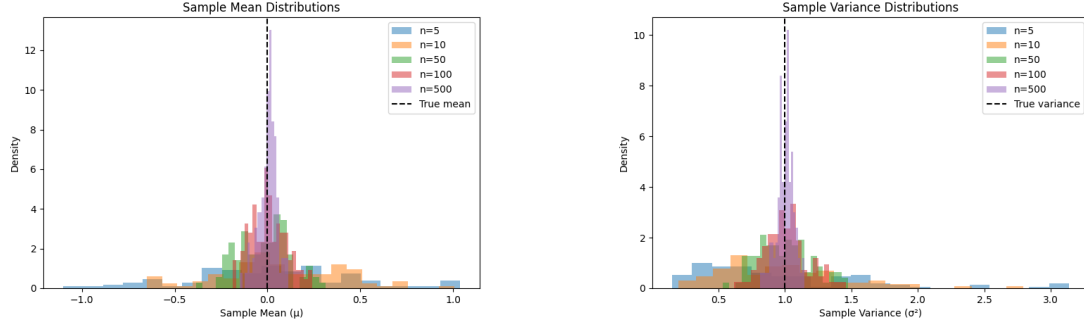
where $\hat{\sigma}_k^2$ estimates the variance of the reward distribution for arm k .

When the sample size is small, the sample mean and sample variance often fluctuate more significantly due to limited data. As the number of samples increases, the sample mean and variance become more stable, converging to the true population values:

$$\mu_k = \lim_{n_k \rightarrow \infty} \hat{\mu}_k \quad \text{and} \quad \sigma_k^2 = \lim_{n_k \rightarrow \infty} \hat{\sigma}_k^2 \quad (3)$$

Thus, as more samples are collected, the variance estimates improve. This makes exploration focus on areas with more uncertainty. In our algorithm, variance changes are used as an exploration signal. This concept can be further clarified by examining a situation where a bandit arm has been sampled only a limited number of times. Under such conditions, the

sample mean may not reliably approximate the true expected reward, and the corresponding sample variance is typically large, indicating significant uncertainty about the reward distribution of arm. As more samples are collected, the variance lowers and becomes stable. This gives a clearer picture of the arm’s reward characteristics. This stabilization enhances the agent’s confidence in deciding the best arm. To illustrate this, Figure 1a shows the distribution of sample means (Equation 1) for different sample sizes, demonstrating larger fluctuations for smaller n_k . Similarly, Figure 1b presents the distribution of sample variances (Equation 2).



(a) Distribution of sample means ($\hat{\mu}_k$) for a bandit arm with true mean $\mu_k = 0$, computed using Equation 1 across 100 trials for various sample sizes n_k .

(b) Distribution of sample variances ($\hat{\sigma}_k^2$) for a bandit arm with true variance $\sigma_k^2 = 1$, computed using Equation 2 across 100 trials for various sample sizes n_k .

Figure 1: Distributions of sample means and variances for a bandit arm

From a statistical point, we can see the relationship between the sample size and the variance. With a smaller sample size, the variance of the samples tends to be higher. This means the reward estimate is less reliable. The reason is not enough data to form a stable estimate. This phenomenon can be explained by the *Law of Large Numbers*. As we get more observations, sample statistics, like mean and variance, get closer to the actual population statistics. [33, 34]

3.2 Decaying Coefficient Design for Exploration Control

We think that the classical Upper Confidence Bound (UCB) approach should add a variance-based exploration term with a decaying coefficient, i.e. $\alpha_t = \alpha_0 / \log(t + \epsilon)$. This matches recent research highlighting its role in effective exploration strategies. Djallel Bouneffouf and Irina Rish.[35] demonstrated that such adaptive exploration strategies with time-dependent decay are effective across healthcare, finance, and recommendation systems, particularly where exploration costs vary across actions. In the RAVEN-UCB algorithm, the exploration coefficient α_t is designed to decay logarithmically over time, specifically as $\alpha_t = \frac{\alpha_0}{\log(t+\epsilon)}$. This

decay helps fix over-exploration problems in standard UCB algorithms. The exploration term $\sqrt{\frac{\ln t}{N(k)}}$ can lead to persistent sampling of suboptimal arms even after sufficient information has been gathered. By reducing α_t over time, the algorithm focuses from exploration to exploitation, enhancing convergence speed and stability. This is particularly beneficial in non-stationary environments, where reward distributions may change over time. The decaying term lets the algorithm adjust to changes better, balancing exploring new choices and using good ones. Studies show that handling changes with techniques like decaying exploration or sliding windows improves bandit algorithms in dynamic settings.

3.3 Recursive Calculation of Sample Mean and Variance

In multi-armed bandit experiments, as the number of samples increases, the sample mean (Equation 1) and sample variance (Equation 2) change continuously. Traditionally, recalculating these statistics means storing and processing all past rewards for each arm at every step. This needs a lot of computational power and time. This leads to a time complexity of $\mathcal{O}(n)$, where n is the number of samples. This approach becomes inefficient when computational efficiency is paramount.

The sample mean and variance naturally use information from all past data. This allows us to refine them step by step without starting over each time. Using this idea, we make recursive formulas for the sample mean and variance from a series of random variables. This lets us update them quickly at each step without checking old data again. Here are the recursive formulas:

Proposition 1 (Recursive Formula for Sample Mean and Variance). *Given a sequence of random variables $X_1, X_2, \dots, X_n$, the recursive formulas for the sample mean and variance are given by:*

$$\bar{X}_{n+1} = \bar{X}_n + \frac{X_{n+1} - \bar{X}_n}{n+1} \quad (4)$$

$$S_{n+1}^2 = \left(1 - \frac{1}{n}\right) S_n^2 + (n+1) (\bar{X}_{n+1} - \bar{X}_n)^2 \quad (5)$$

The detailed derivation of these recursive formulas is provided in Appendix A.1.

3.4 RAVEN-UCB Algorithm

The RAVEN-UCB algorithm integrates the recursive calculations of the sample mean and variance to dynamically adjust the exploration-exploitation trade-off. The algorithm is outlined in Algorithm 1, where the scores for each arm are computed based on the sample mean, exploration term, and variance term. As established by Auer et al. in UCB1[14], we add this coefficient(α_0) reduces exploration intensity over time while leveraging marginal

changes in sample variance to guide arm selection, especially when reward distributions vary significantly.

Algorithm 1: RAVEN-UCB Algorithm

```

1 Initialize  $N(k) = 0$ ,  $M(k) = 0$ ,  $S^2(k) = 0$  for  $k = 1$  to  $K$ ;
2 Set parameters:  $\alpha_0$ ,  $\beta_0$ ,  $\epsilon$ ,  $T$ ;
3 for each time step  $t = 1$  to  $T$  do
4   if  $t \leq K$  then
5      $k_t \leftarrow t$ ;
6   else
7     Compute  $\alpha_t = \alpha_0 / \log(t + \epsilon)$ ;
8     Compute scores for each arm  $k$ :
          
$$\text{score}(k) = M(k) + \alpha_t \cdot \sqrt{\frac{\ln(t+1)}{N(k)+1}} + \beta_0 \cdot \sqrt{\frac{S^2(k)}{N(k)+1}} + \epsilon$$

          
$$k_t \leftarrow \arg \max_k \text{score}(k)$$

9   Obtain reward  $R_t \sim \text{Distribution}(\mu_{k_t}, \sigma_{k_t})$ ;
10  Update  $N(k_t) \leftarrow N(k_t) + 1$ ;
11   $n \leftarrow N(k_t)$ ;
12   $M(k_t) \leftarrow M(k_t) + \frac{R_t - M(k_t)}{n}$ ;
13  if  $n > 1$  then
14     $S^2(k_t) \leftarrow S^2(k_t) + (R_t - M(k_{t-1})) \cdot (R_t - M(k_t))$ ;
15     $S^2(k_t) \leftarrow \frac{S^2(k_t)}{n-1}$ ;
16  else
17     $S^2(k_t) \leftarrow 0$ ;
18 Output total reward and regret;
```

We have theoretically derived the regret bounds of the RAVEN-UCB algorithm (proof see Appendix A.2).

• **Gap-dependent regret:**

$$R(T) = \mathcal{O} \left(\frac{K \sigma_{\max}^2 \log T}{\Delta} \right), \quad (6)$$

• **Gap-independent regret:**

$$R(T) = \mathcal{O} \left(\sqrt{KT \log T} \right). \quad (7)$$

We summarize the theoretical regret bounds of RAVEN-UCB and several baseline algorithms in Table 2.

Table 2: Comparison of Regret Upper Bounds for Different Algorithms

Algorithm	Gap-Dependent	Gap-Independent
RAVEN-UCB (ours)	$\mathcal{O}\left(\frac{K\sigma_{\max}^2 \log T}{\Delta}\right)$	$\mathcal{O}(\sqrt{KT \log T})$
UCB1[14]	$\mathcal{O}\left(\frac{K \log T}{\Delta}\right)$	$\mathcal{O}(\sqrt{KT \log T})$
UCBV[12]	$\mathcal{O}\left(\sum_{k \neq k^*} \left(\frac{\sigma_k^2 \log T}{\Delta_k} + \log T\right)\right)$	$\mathcal{O}(\sqrt{KT \log T})$

4 Experiments

In this section, we evaluate the performance of the RAVEN-UCB algorithm through three experiments. First, we analyze its regret performance under sub-Gaussian rewards and compare it with classical bandit algorithms to see if our theoretical proofs hold. Next, we conduct a parameter selection study to assess the impact of key parameters and present guide for practical values recommendation. Finally, we compare RAVEN-UCB with advanced algorithms in a simulated logistics scenario. It demonstrates its effectiveness in minimizing regret under dynamic conditions.

4.1 Regret Experiment

To check the theoretical properties of the proposed RAVEN-UCB algorithm, we conducted extensive simulations comparing its performance with the classical UCB1 baseline. Our experiments address two key questions:

- (1) How does variance improve regret performance compared to UCB1 ?
- (2) How does the regret reduction scale with T ?

We set $K = 10$ arms with Bernoulli rewards, where the true means θ_k are drawn uniformly from $[0.8, 0.95]$. UCB1 serves as our baseline algorithm. We measure normalized regret reduction:

$$\frac{R_{\text{UCB1}} - R_{\text{V-UCB}}}{R_{\text{UCB1}}} \times 100\% \quad (8)$$

Hyperparameters $(\alpha_0, \beta_0, \epsilon)$ for RAVEN-UCB are tuned via Optuna [36] over $M = 50$ trials per configuration, with search ranges $\alpha_0 \in [0.01, 10]$, $\beta_0 \in [0.01, 10]$, and $\epsilon \in [10^{-3}, 0.5]$.

Figure 2 shows how regret reduction changes as T increases. As T grows large (e.g. $T \approx 5000$), the empirical variance estimates in RAVEN-UCB stabilize. This enables the algorithm to concentrate exploration on genuinely uncertain arms. Empirically, we observe regret reduction is all above 80%, matching the theoretical gap-dependent bound.

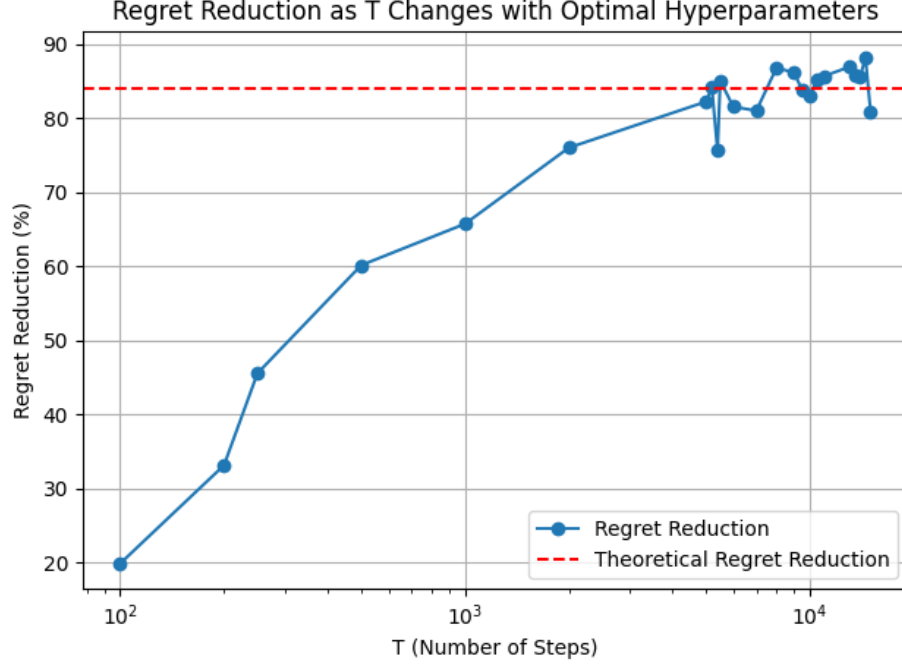


Figure 2: Regret reduction in horizon T

Both algorithms pull each suboptimal arm i on the order of $\log T$ times, but with different scaling factors dependent on reward variance: UCB1 selects suboptimal arms with a rate proportional to $\frac{\log T}{\Delta_i^2}$, our RAVEN-UCB reduces this rate to $\frac{\sigma_i^2 \log T}{\Delta_i^2}$ by incorporating variance exploration (See (32) in A.2). The difference in their pull counts is therefore:

$$\left(\frac{\log T}{\Delta_i^2} - \frac{\sigma_i^2 \log T}{\Delta_i^2} \right) = \frac{(1 - \sigma_i^2) \log T}{\Delta_i^2}, \quad (9)$$

which grows linearly in $\log T$. As T increases, this gap in "mistaken" pulls widens, leading to improved relative regret reduction. The asymptotic improvement ratio converges to $1 - \sigma_{\max}^2$. In our Bernoulli experiment, arm variances $p(1 - p)$ for $p \sim \text{Uniform}(0.8, 0.95)$ lie in $[0.05, 0.16]$, giving $\sigma_{\max}^2 \approx 0.16$, so the regret reduction is approximately $1 - \sigma_{\max}^2 = 0.84$, i.e., 84% regret reduction, which aligns with both theory and empirical results in Figure 2.

4.2 Parameter Selection

Real-world environments are complex and may not match ideal non-stationarity types. Practitioners can use these parameter ranges as a starting point and employ automated hyperparameter optimization techniques, such as Hyperband [37] or bandit-based optimization [38], to fine-tune α_0 and β_0 based on observed performance [39].

We choose three non-stationarity types from Table 1 to conduct experiments and offer

practical guidelines for deploying our algorithm in real-world non-stationary MAB scenarios. The parameter search results for the RAVEN-UCB algorithm, visualized in Figure 3 through cumulative regret curves across α_0 values with varying β_0 , demonstrate consistent hyperparameter selection patterns across distinct non-stationary regimes and time horizons ($T = 1000, T = 10000$).

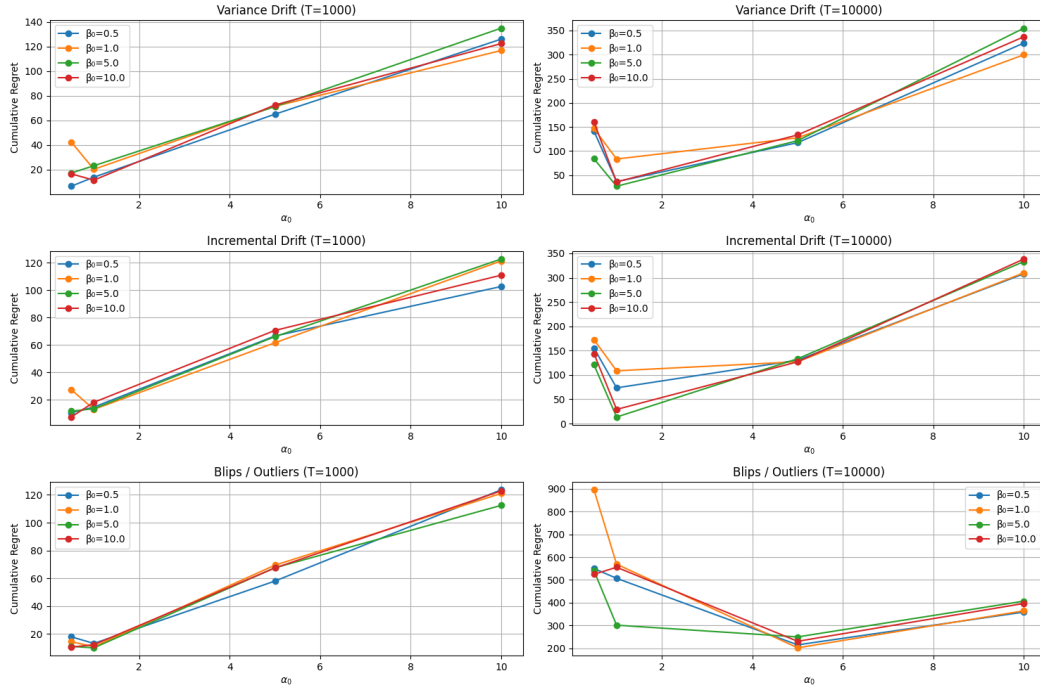


Figure 3: Cumulative Regret versus α_0 for different β_0 values

In the **Variance Drift** scenario, optimal parameters transition from $(\alpha_0 = 0.5, \beta_0 = 0.5)$ with regret 6.55 at $T = 1000$ to $(\alpha_0 = 1.0, \beta_0 = 5.0)$ with regret 27.14 at $T = 10000$. The flat minimal curve for $\beta_0 = 0.5$ at $T = 1000$ indicates conservative exploration in short-term high-variance settings, while the sharp regret decline for $\beta_0 = 5.0$ at $T = 10000$ validates enhanced stability through sustained exploration. For **Incremental Drift**, the $T = 1000$ optimum $(\alpha_0 = 0.5, \beta_0 = 10.0)$ yields regret 7.55 with downward-trending $\beta_0 = 10.0$ curves, contrasting with the balanced $(\alpha_0 = 1.0, \beta_0 = 5.0)$ configuration (regret 13.48) dominating at $T = 10000$. In **Blips/Outliers** environments, short-term optimality at $T = 1000$ is achieved with $(\alpha_0 = 1.0, \beta_0 = 5.0)$ (regret 9.95), whereas long-term performance peaks at $T = 10000$ with $(\alpha_0 = 5.0, \beta_0 = 1.0)$ (regret 202.37), where the $\beta_0 = 1.0$ curve dips at elevated α_0 values.

Observing Figure 3, a clear pattern for choosing hyperparameters appear: (1) α_0 scales with time horizon length, increasing from 0.5 (short-term) to 5.0 (long-term) to balance exploration duration; (2) β_0 inversely correlates with environmental volatility, with lower values (0.5–1.0) stabilizing high-variance regimes and higher values (5.0–10.0) smoothing

gradual drifts. This structured parameter adaptation is very different from passive methods [9] like Discounted UCB and Sliding Window UCB. These require manual tuning of discount factors or window sizes across scenarios. RAVEN-UCB’s mechanism can reduce hyperparameter sensitivity. This is shown by the consistent regret trends across β_0 values for each non-stationarity type.

In high-variance scenarios like traffic delay management, where delays vary due to peak hours or incidents, the preference for $\beta_0 = 5.0$ with $\alpha_0 = 1.0$ suggests that focusing on variance control over extended periods optimizes route selection, minimizing regret. For gradual changes, such as user preference shifts in recommendation systems, the optimal $\alpha_0 = 1.0$, $\beta_0 = 5.0$ supports a good strategy at $T = 10000$, adjustable to $\alpha_0 = 0.5$, $\beta_0 = 10.0$ for short-term (e.g., daily) recommendations to quickly adapt to new preferences. In long-tail reward scenarios, such as inventory management during promotional spikes, the shift to $\alpha_0 = 5.0$, $\beta_0 = 1.0$ emphasizes aggressive exploration to capture rare high rewards without over-penalizing variance, aligning with the plot’s trend for low β_0 .

4.3 Simulation Study—Logistics Optimization Scenario

In modern logistics and supply chain management, companies must make decisions under uncertainty. This makes the environment unpredictable. Traditional decision-making algorithms often struggle in such settings. So, companies need adaptive methods to quickly respond to changes. This scenario is highly relevant to several domains within logistics and supply chain systems. Logistics robotics and port operations require adaptive algorithms to manage dynamic operational constraints and demand surges [40]. We can model these scenarios as a non-stationary multi-armed bandit problem. Here, each warehouse is an arm, and the reward is the delivery efficiency score.

We model a logistics optimization scenario with K warehouses (arms), where the efficiency score of assigning an order to a warehouse is drawn from a normal distribution $\mathcal{N}(\mu, \sigma^2)$. The range of means $[\mu_{\min}, \mu_{\max}]$ represents realistic variations in warehouse performance, where higher values indicate faster and more cost-effective operations. The variance range $[\sigma_{\min}^2, \sigma_{\max}^2]$ captures the uncertainty inherent in logistics, such as unpredictable delays or variable processing times. Every R time steps (representing customer orders), the means and variances of approximately one-third of the warehouses ($\frac{K}{3}$ arms) are randomly reset to new values within their respective ranges using a uniform distribution.

Table 3: Simulation parameters and values used in the experiment

Symbol	Description	Value used in experiment
K	Warehouses	100
$\mu_{\min}, \mu_{\max}$	Range of mean efficiency scores	0.3, 0.8
$\sigma_{\min}^2, \sigma_{\max}^2$	Range of variances	0.01, 0.09
R	Shift interval	5,000
T	Time steps	50,000
N	Number of independent trials	50

This setup mimics real-world operational changes, such as traffic delays, inventory restocking, or weather-induced disruptions, which are common in logistics systems [41]. What’s more, peak traffic hours, inventory updates, or seasonal weather patterns will also result in these situations, which are critical challenges in logistics optimization [42]. The simulation runs for T time steps, with N independent trials to ensure statistical reliability, and uses a fixed random seed for reproducibility.

By comparing RAVEN-UCB with eight other established MAB algorithms (UCB [14], UCB-V [12], ϵ -greedy [43], Thompson Sampling [44], f-dsw TS (min) [11], WLS + Optimistic TS [45], CCB [1], and UCB-Imp [46]) in this logistics-inspired setting, we aim to demonstrate its effectiveness in dynamic decision-making environments. This research builds upon prior work in applying MAB algorithms to logistics optimization [28] and extends it by focusing on non-stationary environments [47].

Table 4: Simulation Results

Algorithm	Cumulative Reward	Cumulative Regret	Suboptimal Pulls (per trial)
RAVEN-UCB	38276.8	1717.8	40824.2
UCB	34551.2	5446.1	46595.1
UCB-V	33985.8	6011.4	46475.4
ϵ -greedy	36595.1	3397.8	43147.7
Thompson Sampling	37029.8	2956.9	46254.5
f-dsw TS (min)	34701.3	5305.7	48003.3
WLS + Optimistic TS	36224.0	3771.3	46357.8
CCB	37306.9	2690.5	41095.1
UCB-Imp	28192.8	11798.9	49446.5

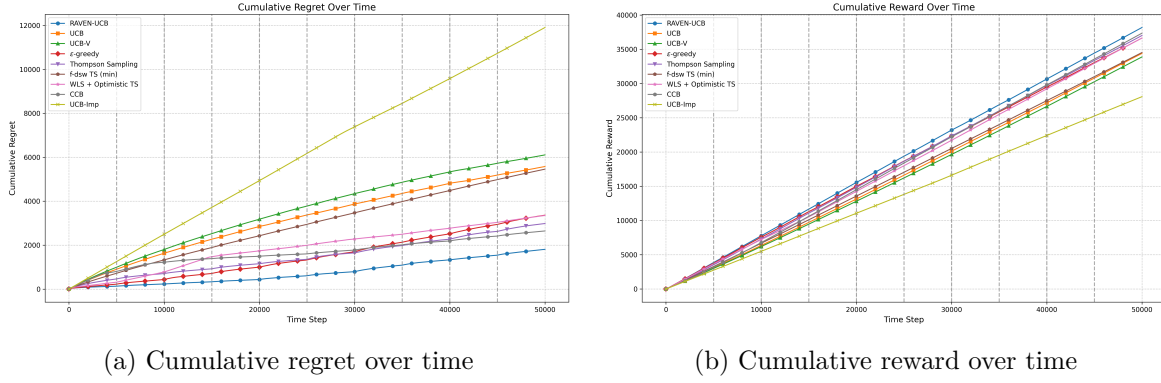


Figure 4: (a) Cumulative regret, (b) cumulative reward

The results, averaged over 50 independent trials, are in Table 4. RAVEN-UCB achieves the highest cumulative reward of 38,276.8 and the lowest cumulative regret of 1,717.8 among all compared algorithms. The next best algorithm, CCB, records a cumulative reward of 37,306.9 and a regret of 2,690.5. In contrast, the standard UCB algorithm exhibits a significantly higher regret of 5,446.1. See Figure 4a and Figure 4b.

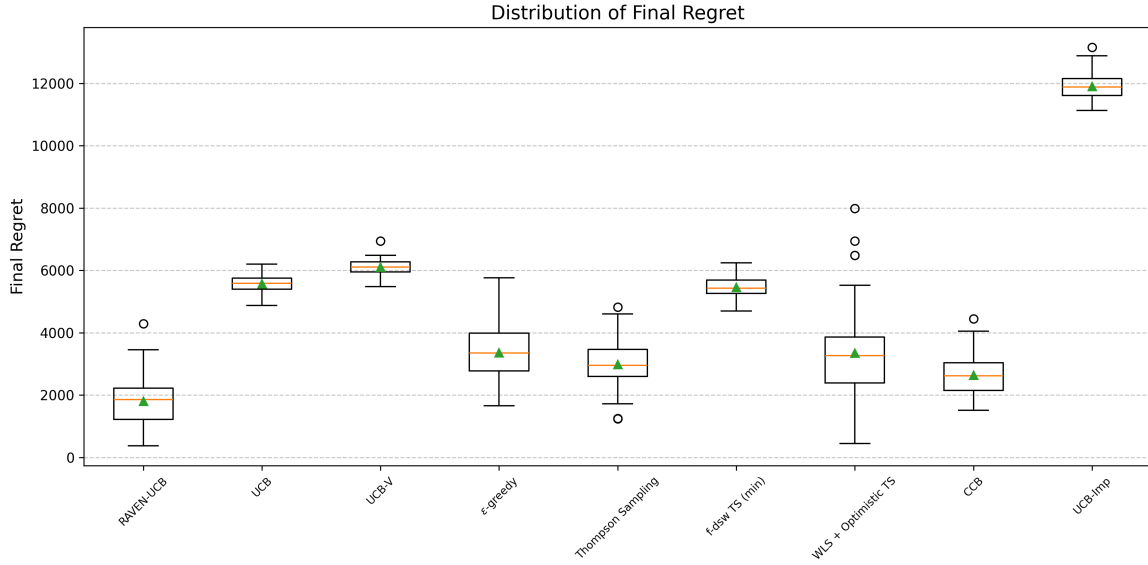


Figure 5: Boxplots of the results

RAVEN-UCB performs well because it uses a variance-adaptive exploration strategy. This helps it quickly find and adapt to changes in warehouse efficiency. This is particularly valuable in logistics, where rapid adaptation to operational changes is critical. But using a normal distribution for rewards fits continuous efficiency metrics. It may not be the best choice where binary outcomes (e.g., on-time delivery rates) are more relevant.

5 Conclusion

In this paper, we introduced and analyzed the RAVEN-UCB algorithm. It addresses non-stationarity in Multi-Armed Bandit (MAB) problems through a variance-adaptive approach. This dynamically balances exploration and exploitation. We review the MAB framework and its uses in online advertising, recommendation systems, and logistics optimization. We classifying non-stationarity into distributional parameter changes (DPC), periodic changes (PC), and temporary fluctuations (TF). The RAVEN-UCB algorithm uses a new approach with a variance-based exploration signal. It uses the sample variance to guide exploration, and employs a logarithmically decaying coefficient $\alpha_t = \frac{\alpha_0}{\log(t+\epsilon)}$ to adjust exploration intensity. Computational efficiency improves with recursive formulas, reducing the time complexity to $\mathcal{O}(1)$. Theoretically, the algorithm achieves gap-dependent regret $\mathcal{O}\left(\frac{K\sigma_{\max}^2 \log T}{\Delta}\right)$ and gap-independent regret $\mathcal{O}(\sqrt{KT \log T})$. We conducted many experiments to evaluate the RAVEN-UCB algorithm:

Firstly, in regret performance experiments with sub-Gaussian rewards, RAVEN-UCB achieved $1 - \sigma_{\max}^2$ regret reduction with Bernoulli rewards. This matches its gap-dependent regret bound. It outperforms UCB1's $\mathcal{O}\left(\frac{K \log T}{\Delta}\right)$ much better.

Secondly, in the parameter selection experiments across three representative non-stationary scenarios, grid search over α_0 and β_0 helped with practical tuning strategies. These findings provide actionable guidance for using RAVEN-UCB in dynamic environments and support stable performance across varying types of non-stationarity.

Thirdly, in a simulated logistics optimization scenario with $K = 100$ warehouses as arms and rewards drawn from a normal distribution $\mathcal{N}(\mu, \sigma^2)$, RAVEN-UCB recorded a cumulative regret of 1717.8 over $T = 50,000$ steps and $N = 50$ trials, far surpassing UCB1's 5446.1, demonstrating its ability to adapt to changing efficiency distributions via a variance-adaptive strategy.

Based on our experiments, we provide the following practical guidance for parameter selection in RAVEN-UCB:

- (1) In high-variance environments (e.g., traffic delay management), setting $\alpha_0 = 1.0$, $\beta_0 = 5.0$ helps stabilize exploration over long horizons.
- (2) For gradual changes (e.g., evolving user preferences), $\alpha_0 = 1.0$, $\beta_0 = 5.0$ is effective for long-term adaptation, while $\alpha_0 = 0.5$, $\beta_0 = 10.0$ works better for short-term responsiveness.
- (3) In rare-event or outlier-driven contexts (e.g., promotional spikes in inventory management), aggressive exploration with $\alpha_0 = 5.0$, $\beta_0 = 1.0$ captures high-reward opportunities efficiently.

Future work could explore applying RAVEN-UCB to real-world datasets in domains such as e-commerce, traffic management, or financial decision-making, where non-stationarity is inherent and variance dynamics are complex. This would help validate its robustness and

scalability beyond controlled simulations. Additionally, another direction is to investigate automated adaptation mechanisms for hyperparameter tuning, potentially leveraging meta-learning or reinforcement meta-bandits. Also, integrating RAVEN-UCB with contextual bandit frameworks to incorporate side information may boost its performance in diverse environments.

Declarations

- Data availability: This study does not use external datasets. All data were generated through simulations as described in Section 4. The data can be made available upon reasonable request from the corresponding author.
- Code availability: The code used to implement the RAVEN-UCB algorithm and conduct the simulations in Sections 4 is available at <https://github.com/66661654/Raven-UCB>.
- Author contribution: Conceptualization, J. Fang and Yuxun Chen; Methodology, J. Fang and Yuxun Chen; Coding, J. Fang; Validation, Yuxun Chen; Writing—original draft, J. Fang and Yuxun Chen; Writing—review and editing, Yuxun Chen, Yuxin Chen, and C. Zhang. All authors have read and agreed to the published version of the manuscript.

A Appendix

A.1 Derivation of Recursive Formulas for Sample Mean and Variance

Upon the arrival of a new observation x_{n+1} , the sample mean for $n + 1$ observations, $\bar{x}_{n+1}$, is similarly defined:

$$\bar{x}_{n+1} = \frac{1}{n+1} \sum_{i=1}^{n+1} x_i. \quad (10)$$

$$\sum_{i=1}^{n+1} x_i = \sum_{i=1}^n x_i + x_{n+1} = n\bar{x}_n + x_{n+1}. \quad (11)$$

So, we get (4):

$$\bar{x}_{n+1} = \bar{x}_n + \frac{x_{n+1} - \bar{x}_n}{n+1}. \quad (12)$$

Turning to the sample variance, we adopt the unbiased estimator, which accounts for the degrees of freedom lost in estimating the sample mean. For n observations, the sample

variance S_n^2 is defined as:

$$S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x}_n)^2, \quad \text{for } n \geq 2. \quad (13)$$

Let us define $M_n = \sum_{i=1}^n (x_i - \bar{x}_n)^2$, so that

$$S_n^2 = \frac{M_n}{n-1}. \quad (14)$$

$$S_{n+1}^2 = \frac{1}{n} \sum_{i=1}^{n+1} (x_i - \bar{x}_{n+1})^2 = \frac{M_{n+1}}{n}, \quad (15)$$

where

$$M_{n+1} = \sum_{i=1}^{n+1} (x_i - \bar{x}_{n+1})^2. \quad (16)$$

$$M_{n+1} = \sum_{i=1}^n (x_i - \bar{x}_{n+1})^2 + (x_{n+1} - \bar{x}_{n+1})^2. \quad (17)$$

We need to express $x_i - \bar{x}_{n+1} = (x_i - \bar{x}_n) - (\bar{x}_{n+1} - \bar{x}_n)$, noting that $\sum_{i=1}^n (x_i - \bar{x}_n) = 0$. Summing over $i = 1$ to n , of course, using (4):

$$\sum_{i=1}^n (x_i - \bar{x}_{n+1})^2 = M_n + n(\bar{x}_{n+1} - \bar{x}_n)^2. \quad (18)$$

Next, compute the contribution of the new observation,

$$x_{n+1} - \bar{x}_{n+1} = x_{n+1} - \left(\bar{x}_n + \frac{x_{n+1} - \bar{x}_n}{n+1} \right) = \frac{n(x_{n+1} - \bar{x}_n)}{n+1}, \quad (19)$$

so:

$$(x_{n+1} - \bar{x}_{n+1})^2 = \left(\frac{n}{n+1} (x_{n+1} - \bar{x}_n) \right)^2 = \frac{n^2}{(n+1)^2} (x_{n+1} - \bar{x}_n)^2. \quad (20)$$

Additionally,

$$(\bar{x}_{n+1} - \bar{x}_n)^2 = \left(\frac{x_{n+1} - \bar{x}_n}{n+1} \right)^2 = \frac{1}{(n+1)^2} (x_{n+1} - \bar{x}_n)^2. \quad (21)$$

Therefore:

$$M_{n+1} = M_n + n \cdot \frac{1}{(n+1)^2} (x_{n+1} - \bar{x}_n)^2 + \frac{n^2}{(n+1)^2} (x_{n+1} - \bar{x}_n)^2 = M_n + \frac{n}{n+1} (x_{n+1} - \bar{x}_n)^2. \quad (22)$$

Hence, the updated sample variance is:

$$S_{n+1}^2 = \frac{M_{n+1}}{n} = \frac{M_n}{n} + \frac{1}{n+1}(x_{n+1} - \bar{x}_n)^2. \quad (23)$$

Since $M_n = (n-1)S_n^2$, we have $\frac{M_n}{n} = \frac{(n-1)S_n^2}{n} = \left(1 - \frac{1}{n}\right)S_n^2$, and recognizing that $(x_{n+1} - \bar{x}_n)^2 = (n+1)^2(\bar{x}_{n+1} - \bar{x}_n)^2$, the variance update can be expressed as (5):

$$S_{n+1}^2 = \left(1 - \frac{1}{n}\right)S_n^2 + (n+1)(\bar{x}_{n+1} - \bar{x}_n)^2. \quad (24)$$

Each operation executes in constant time, independent of n , yielding a time complexity of $O(1)$ per update. In contrast, the naive approach recomputes the mean and variance from all $n+1$ observations, requiring $O(n)$ time per update, with a total complexity of $O(n^2)$ over n updates.

A.2 Proof of the Regret Upper Bound

Consider a multi-armed bandit with K arms. Each arm k has rewards X_k with mean μ_k and sub-Gaussian parameter σ_k^2 . The optimal arm is $k^* = \arg \max_k \mu_k$ with mean $\mu^* = \mu_{k^*}$. The gap for a suboptimal arm $k \neq k^*$ is $\Delta_k = \mu^* - \mu_k > 0$, and $\Delta = \min_{k \neq k^*} \Delta_k$. The cumulative regret over T rounds is :[14]:

$$R(T) = \mathbb{E} \left[\sum_{t=1}^T (\mu^* - \mu_{k_t}) \right] = \sum_{k \neq k^*} \Delta_k \cdot \mathbb{E}[N_k(T)], \quad (25)$$

where $N_k(T)$ is the number of pulls of arm k .

The upper confidence bound $U_k(t)$ typically includes the empirical mean $\hat{\mu}_k(t)$ and an exploration term. For simplicity, we define $U_k(t) = \hat{\mu}_k(t) + c_k(t)$, where the confidence radius $c_k(t)$ is given in Algorithm 1.

$$c_k(t) = \alpha_t \cdot \sqrt{\frac{\ln(t)}{N_k(t) + 1}} + \beta_0 \cdot \sqrt{\frac{\hat{\sigma}_k^2}{N_k(t) + 1}} + \epsilon, \quad (26)$$

where $\alpha_t = \alpha_0 / \log(t + \epsilon)$. When T is large, the term ϵ becomes negligible. Moreover, the empirical variance $\hat{\sigma}_k^2$ converges to the true variance σ_k^2 . Suppose $\alpha_0 \sqrt{\ln t} + \beta_0 \sigma_k = 2\sigma_k \sqrt{\ln t}$, the confidence radius can be simplified to:

$$c_k(t) = \sqrt{\frac{4\sigma_k^2 \log t}{N_k(t)}}, \quad (27)$$

For sub-Gaussian rewards we use the concentration inequality [48] :

$$\mathbb{P}(|\hat{\mu}_k(t) - \mu_k| \geq \varepsilon) \leq 2 \exp \left(-\frac{N_k(t)\varepsilon^2}{2\sigma_k^2} \right). \quad (28)$$

Define $E_k(t) = \{k_t = k\}$. Then $E_k(t) \subseteq A_t \cup B_t$, where $A_t = \{U_{k^*}(t) < \mu^*\}$, $B_t = \{U_k(t) \geq \mu_k + \frac{\Delta_k}{2}\}$, so:

$$\mathbb{P}(E_k(t)) \leq \mathbb{P}(A_t) + \mathbb{P}(B_t). \quad (29)$$

For A_t :

$$\mathbb{P}(A_t) = \mathbb{P}(\hat{\mu}_{k^*}(t) - \mu^* < -c_{k^*}(t)) \leq \exp \left(-\frac{N_{k^*}(t)c_{k^*}(t)^2}{2\sigma_{k^*}^2} \right) = \frac{1}{t^2}, \quad (30)$$

since $c_{k^*}(t)^2 = \frac{4\sigma_{k^*}^2 \log t}{N_{k^*}(t)}$. For B_t , set $\varepsilon = \frac{\Delta_k}{2} - c_k(t)$. If $N_k(t) \geq \frac{16\sigma_k^2 \log t}{\Delta_k^2}$, then $c_k(t) \leq \frac{\Delta_k}{2}$, and:

$$\mathbb{P}(B_t) = \mathbb{P} \left(\hat{\mu}_k(t) - \mu_k \geq \frac{\Delta_k}{2} - c_k(t) \right) \leq \exp \left(-\frac{N_k(t) \left(\frac{\Delta_k}{2} \right)^2}{2\sigma_k^2} \right) \leq \frac{1}{t^2}. \quad (31)$$

Use (29), $\mathbb{P}(E_k(t)) \leq \frac{2}{t^2}$. Set $n_k = \frac{16\sigma_k^2 \log T}{\Delta_k^2}$. Noting the result of Basel Problem [49, 50], the expected sub-pulls are : $\mathbb{E}[N_k(T)] \leq n_k + \sum_{t=n_k+1}^T \frac{2}{t^2} < n_k + \frac{\pi^2}{6} = \mathcal{O} \left(\frac{\sigma_k^2 \log T}{\Delta_k^2} \right)$.

The gap-dependent regret is:

$$\mathbb{E}[R(T)] < \sum_{k \neq k^*} \frac{16\sigma_k^2 \log T}{\Delta_k^2} \leq \mathcal{O} \left(\frac{K\sigma_{\max}^2 \log T}{\Delta} \right). \quad (32)$$

For the gap-independent bound, assume $\mathbb{E}[N_k(T)] \approx l$, with:

$$c_k(t) = \sqrt{\frac{4\sigma_k^2 \log T}{l}}, \quad \Delta_k \approx c_k(t). \quad (33)$$

By Cauchy-Schwarz on (25):

$$\mathbb{E}[R(T)] \leq \sqrt{(K-1)\Delta_k^2} \cdot \sqrt{(K-1)l^2}. \quad (34)$$

Substitute (33):

$$\mathbb{E}[R(T)] \approx \sqrt{4\sigma_k^2(K-1) \log T \cdot l}. \quad (35)$$

With $\sum_{k=1}^K N_k(T) = T$, set $m = (K-1)l \approx \sqrt{KT \log T}$, so:

$$l \approx \sqrt{\frac{T \log T}{K}}, \quad \mathbb{E}[R(T)] \approx \sqrt{4\sigma_{\max}^2 K T \log T} = \mathcal{O}(\sqrt{KT \log T}). \quad (36)$$

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Sentiment-Driven Nonlinear Relationship Between Financial Market Emotion and Returns: Dynamic Adaptive Modeling Based on LLMs

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Abstract—This study investigates the nonlinear relationship between investor sentiment and market returns in two Chinese financial markets, emphasizing asymmetric effects across bull and bear regimes. Via a fine-tuned Chinese RoBERTa model, we analyze over financial news texts everyday to generate sentiment scores. Two novel indices, the Momentum-Liquidity Index (MLI) and Price Stability Adjusted Equal-Weight Index (PSAEWI), capture short-term volatility and long-term trends. Empirical results, validated through VAR, quantile regression, and XGBoost, confirm sentiment’s stronger predictive power in bull markets and its state-dependent interactions, and we further use them in prediction of macro-performance indicator. Also, we propose a real-time monitoring system and a dual-model framework integrates sentiment classification and emotion regression, dynamically weighted by market conditions. Finally, we offer practical applications for investors and regulators in behavioral finance.

Index Terms—Sentiment Analysis, Financial Markets, LLMs, Nonlinear Modeling

I. INTRODUCTION

Investor sentiment plays a vital role in shaping financial market dynamics, a cornerstone of behavioral finance that elucidates how psychological factors can lead to asset mispricing [1]. In the Chinese financial market, characterized by a predominance of retail investors, sentiment-driven effects are particularly pronounced, often amplifying market volatility and creating unique challenges for asset pricing models [2]. Traditional sentiment measurement techniques, such as dictionary-based methods, are limited by their inability to capture contextual nuances and adapt to the domain-specific language of financial texts.

Recent advancements in natural language processing, particularly the development of large language models (LLMs) such as BERT, have demonstrated transformative potential in understanding intricate semantics within financial texts [3]. Domain-specific adaptations, including FinBERT and BloombergGPT, further highlight the ability of LLMs to decipher nuanced financial narratives, offering a significant leap over traditional methods [4, 5]. Despite these technological strides, significant research gaps persist. Current studies often fail to adequately model the nonlinear dynamics between sentiment and returns,

particularly in distinguishing the differential impacts across bull and bear markets [6]. Moreover, the dynamic nature of financial markets necessitates adaptive mechanisms that can respond promptly to evolving conditions, a challenge that existing sentiment monitoring systems struggle to meet due to technical bottlenecks [7].

To address these gaps, we use a framework that leverages LLMs to generate high-quality sentiment labels and integrates them with two innovative market indices—the Momentum-Liquidity Index (MLI) and the Price Stability Adjusted Equal-Weight Index (PSAEWI)—to capture nonlinear and state-dependent effects on market returns. The core of our approach is a dual-model fusion architecture that combines a three-label sentiment polarity classifier with a seven-emotion intensity regressor. By doing this, we can weighted based on prevailing market conditions dynamically.

Our research contributes by empirically validating the asymmetric transmission mechanism of sentiment on market returns, demonstrating that sentiment effects are more pronounced in bull markets than in bear markets. It also establishes state-dependent stochastic process models that account for market heterogeneity, offering a new perspective on how sentiment interacts with market conditions. Empirically, our findings suggest that LLM-derived sentiment factors are significant predictors of macroeconomic indicators, such as the Purchasing Managers’ Index (PMI) and financing growth, and reveal differential sentiment sensitivities between the ChiNext and main board markets. From a practical standpoint, we propose a real-time risk warning system that utilizes Apache Kafka and Spark for streaming data processing, SHAP for model interpretability, and an isolation forest algorithm for anomaly detection. This system offers a robust toolkit for investors to optimize trading strategies and for regulators to monitor market stability, addressing the technical bottlenecks identified in prior work [7].

We organize the remainder of this paper as follows: Section II surveys financial sentiment measurement techniques, nonlinear market relationships, and real-time monitoring systems. Section III introduces our dual-model sentiment framework

with dynamic adaptive weighting (DAO module) and constructs two novel market indices (MLI and PSAEWI). Section IV details the Chinese financial text corpus, market datasets, and preprocessing protocols. Section V empirically validates four hypotheses through interaction regressions and machine learning models across bull/bear regimes. The concluding section (VI) synthesizes key findings and discusses policy applications for financial institutions and regulators.

II. LITERATURE REVIEW

A. Traditional Sentiment Measurement Methods

Sentiment analysis in finance began with dictionary-based methods, which use predefined word lists to classify text sentiment [8]. For instance, the Loughran-McDonald dictionary, designed for financial texts, is widely used in research [8]. However, dictionary-based methods fail to capture context, sarcasm, or subtle language differences, limiting their predictive accuracy [9]. Tetlock (2007) found that negative words in news reports predict lower market returns, while positive words have insignificant effects [9].

B. Application of Large Language Models in Finance

The advent of large language models (LLMs) marks a significant breakthrough in natural language processing. BERT [3], with its bidirectional self-attention mechanism, captures contextual information effectively, making it particularly suitable for tasks like financial sentiment analysis. Finance-specific LLMs, such as FinBERT [4] and BloombergGPT [5], fine-tuned on financial texts, have significantly improved classification performance. Li et al. (2023) reviewed LLMs' applications in finance, highlighting their potential in trading, risk modeling, and sentiment analysis while noting challenges in bias and interpretability [10]. Yang et al. (2023) introduced retrieval-augmented generation (RAG) to enhance sentiment analysis by incorporating external knowledge bases [11].

In the Chinese market, Jiang et al. (2023) developed a Chinese financial LLM by fine-tuning BERT with market returns, demonstrating its superiority over traditional dictionary-based methods in predicting market returns and risks, especially during extreme events [2]. Additionally, Jiang et al. (2024) compared OPT, BERT, and FinBERT on U.S. financial news, finding OPT achieved the highest accuracy of 74.4% in predicting stock returns [12].

C. Nonlinear Relationship Between Sentiment and Market Returns

Research indicates that the relationship between sentiment and market returns is nonlinear. Baker and Wurgler (2006) found that sentiment has a stronger impact on small, young, and high-volatility stocks [13]. Wang et al. (2016) used quantile regression to show that sentiment has a greater positive effect on high-return stocks and a lesser effect on low-return stocks [14]. Zhang and Wang (2019) further confirmed the differential effects of sentiment in bull and bear markets [6]. In the Chinese market, Jiang et al. (2019) and Liu et al.

(2020) found that sentiment's impact on stock returns is more pronounced during market downturns [15, 16].

Bull and bear markets represent typical market states where sentiment's impact differs. Antweiler and Frank (2004) found that internet stock forum sentiment is more informative during bear markets [17]. Das and Chen (2007) used Yahoo! Finance message board data to show that sentiment can predict stock returns, with stronger relationships during high-volatility periods [18]. In the Chinese market, Zhang and Wang (2019) found that sentiment's impact on stock returns is more significant during bear markets [6].

D. Real-Time Sentiment Monitoring Systems

With the real-time availability of social media and news data, real-time sentiment monitoring systems have become a research focus. Feldman et al. (2010) developed a system based on general knowledge rules to predict stock prices using news sentiment [7]. In recent years, LLMs have been used to build real-time sentiment analysis systems, leveraging their ability to process large volumes of text data quickly [10]. For example, streaming data processing frameworks based on Apache Kafka and Spark can generate real-time sentiment scores and visualize them through Dash dashboards [19].

III. METHODOLOGY

A. Definition of Bull and Bear Markets

To analyze the differential impact of sentiment across market states, we first establish a systematic methodology for identifying bull and bear markets. Our classification approach combines technical indicators with market dynamics, providing a robust framework for market state identification. We define bull and bear markets using a 250-day moving average (MA) as the primary technical indicator, [20] supplemented by relative price deviation metrics:

- **Bull Market:** Periods when the closing price exceeds its 250-day MA
- **Bear Market:** Periods when the closing price falls below its 250-day MA

Mathematically, the market state indicator I_t for any trading day t is defined as:

$$I_t = \begin{cases} 1 \text{ (Bull)} & \text{if } P_t > \text{MA}_{250,t} \\ 0 \text{ (Bear)} & \text{if } P_t \leq \text{MA}_{250,t} \end{cases}$$

where P_t is the closing price and $\text{MA}_{250,t}$ represents the 250-day moving average.

B. New Market Index

We propose two novel indices: the Momentum-Liquidity Index (MLI) and the Price Stability Adjusted Equal-Weight Index (PSAEWI), capturing complementary market features.

The MLI integrates volume change, momentum, and volatility to reflect daily market dynamics beyond traditional market-cap weights [21, 22]. Volume change, momentum, and volatility are combined to calculate stock weights as:

$$w_{i,t-1} = \frac{\exp(\alpha \Delta V_{i,t-1} + \beta M_{i,t-1} + \gamma \sigma_{i,t-1})}{\sum_{j=1}^N \exp(\alpha \Delta V_{j,t-1} + \beta M_{j,t-1} + \gamma \sigma_{j,t-1})}, \quad (1)$$

and the index value is

$$\text{MLI}_t = \sum_{i=1}^N w_{i,t-1} (\Delta V_{i,t} + M_{i,t} + \sigma_{i,t}). \quad (2)$$

This approach captures liquidity and momentum effects relevant for risk premia [23, 24].

The PSAEWI emphasizes price stability by weighting stocks according to their closing price relative to the 250-day moving average [21, 25]:

$$\text{ratio}_i = \frac{\text{Closing Price}_i}{250\text{-Day MA}_i}, \quad (3)$$

with weights

$$w_i = \frac{\exp(-k(\text{ratio}_i - 1)^2)}{\sum_{j=1}^N \exp(-k(\text{ratio}_j - 1)^2)}, \quad (4)$$

and the daily index computed as

$$\text{PSAEWI}_t = \sum_{i=1}^N w_i \cdot \frac{1}{\text{ratio}_i}. \quad (5)$$

This index highlights firms with stable pricing, complementing sentiment-based macroeconomic indicators [26, 27].

Their dynamic interplay, as shown in the rolling correlation plot (Figure 1), supports their joint use with sentiment indicators in regression models, offering valuable insights for academic research and practical market-economy studies.

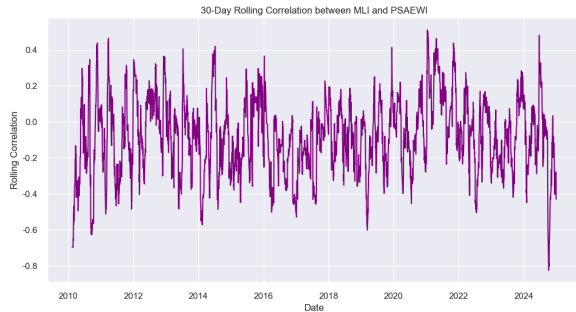


Fig. 1. Rolling correlation between MLI and PSAEWI

C. Sentiment Score Construction

We propose a dual-model framework that integrates categorical sentiment classification and continuous emotional intensity regression, tailored for financial text analysis. The pipeline employs domain-adapted fine-tuning for two specialized models, as illustrated in Figure 2.

We fine-tune the *hfl/chinese-roberta-wwm-ext* model [28] on a stratified dataset of 30,000 annotated financial texts.

Fine-tuning uses focal loss to prioritize misclassified cases, L2 regularization ($\lambda = 0.01$), and cosine annealing (learning rate from 5×10^{-5} to 1×10^{-6}). Hard sample mining ensures learning of context-dependent cases, such as positive labeling of “profit declined but exceeded expectations.”

The *iic/nlp_structbert_emotion-classification_chinese-base* model quantifies seven emotional dimensions, mapped to a continuous polarity score using market-impact-validated weights:

$$s_{\text{reg}} = 1.0 \cdot p_{\text{Joy}} + 0.8 \cdot p_{\text{Like}} - (1.5 \cdot p_{\text{Fear}} + 1.2 \cdot p_{\text{Anger}} + 0.5 \cdot p_{\text{Sadness}}) \quad (6)$$

Neutral emotions (Surprise, Disgust) have weights ≤ 0.3 to minimize noise. Fine-tuning prioritizes forward-looking guidance over historical reports and disambiguates sector-specific terms (e.g., “margin compression” in technology vs. energy).

The final sentiment score integrates outputs from the regression and classification heads to balance *intensity granularity* and *categorical consistency*:

$$s_{\text{final}} = w \cdot s_{\text{reg}} + (1 - w) \cdot (p_{\text{pos}} - p_{\text{neg}}) \quad (7)$$

The default weight ratio $w = 0.7$ (regression head) and $1 - w = 0.3$ (classification head) is optimized on historical earnings surprise events, where nuanced emotional intensity demonstrates stronger return predictability.

The regression head captures fine-grained emotional shifts (e.g., “moderate optimism” vs. “strong bullishness”), while the classification head ensures directional consistency (e.g., preventing extreme intensity misattribution in neutral contexts).

We sampled 30,000 news text data evenly on a daily basis from the original data sample, and calculated the initial sentiment score for each news text, and obtained the results as shown in Figure 3.

D. Relevance Score Construction

To guarantee that only finance-related news is fed into the downstream models, we design a *relevance score* that quantifies the degree to which each document is associated with financial markets. The construction follows a two-step hybrid procedure that combines a domain-specific lexicon with named-entity recognition (NER), thereby capturing both explicit financial terminology and implicit entity references.

We employ the **CNEconDict** as a benchmark vocabulary. Let d_i denote the i -th news article and $\mathcal{L}$ the lexicon. After Chinese word segmentation, we compute the term-level ratio

$$T_i = \frac{|\{w \in d_i : w \in \mathcal{L}\}|}{|d_i|},$$

where the numerator counts lexicon hits and the denominator is the total word count of d_i . T_i provides a first-pass indication of financial content.

Next, we extract named entities using a BERT-based Chinese NER model. Let E_i be the set of entities identified in d_i . We measure entity overlap with the lexicon as

$$E_i^* = \frac{|E_i \cap \mathcal{L}|}{|E_i|}.$$

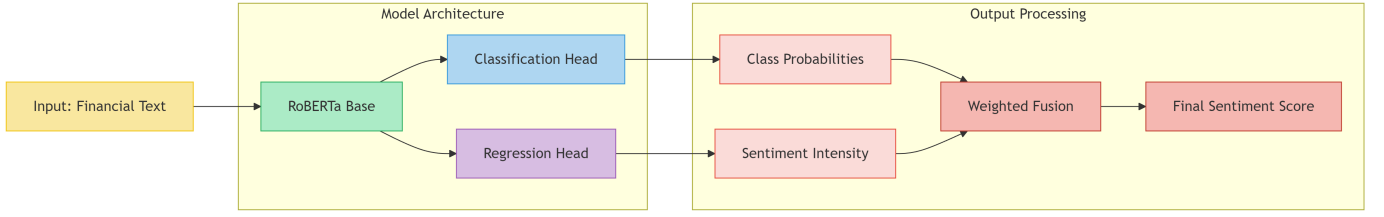


Fig. 2. Sentiment scoring pipeline: (1) 3-category classifier (Negative/Neutral/Positive), (2) 7-emotion regressor, (3) weighted fusion module.

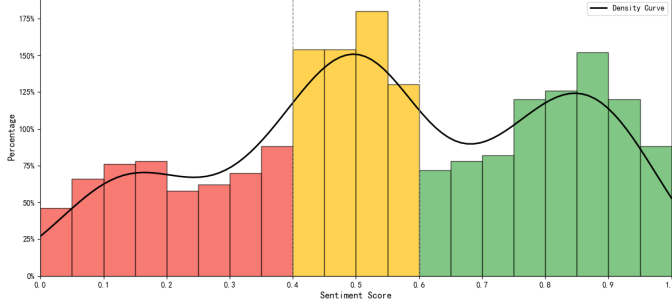


Fig. 3. Bimodal distribution of final sentiment scores: central mode (stable market conditions) and tails (extreme events e.g., 2015 market crash).

For example, if d_i contains 20 entities and 8 of them match the lexicon, then $E_i^* = 0.40$.

The final relevance score R_i is a convex combination of the two components:

$$R_i = \alpha T_i + (1 - \alpha) E_i^*, \quad \alpha \in [0, 1], \quad (8)$$

where we set $\alpha = 0.5$ by default and later test $\alpha \in \{0.25, 0.75\}$ for robustness. Documents with $R_i < 0.10$ are discarded as non-financial noise.

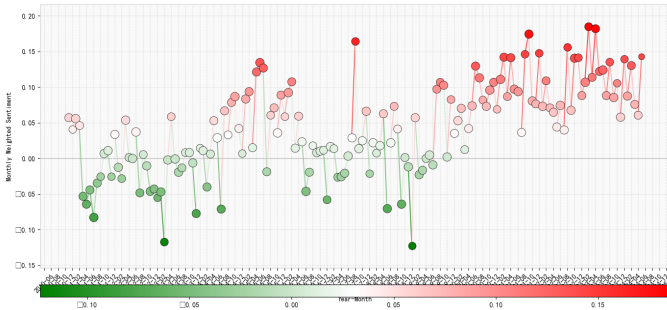


Fig. 4. Time-series plot of changes in initial mood scores per month (2011-2024)

We use Eq. (8) to the raw corpus (January 2015–December 2025) eliminates 18.3% of the news items, leaving a final sample of 2.74 million articles with a mean (median) R_i of 0.46 (0.42). The filtering step significantly improves the downstream sentiment–return correlations.

E. Dynamic Adaptive Optimization Module

1) *Module Design Objectives:* The Dynamic Adaptive Optimization (DAO) module addresses the temporal volatil-

ity characteristic of financial markets by enabling continuous model adaptation. Designed as a parameter optimization framework inspired by [29], it achieves the following objectives:

- **Adaptiveness:** Dynamic recalibration of model parameters in response to market regime shifts (e.g., bull-to-bear transitions)
- **Computation Efficiency:** 63-82% reduction in computational overhead compared to full model retraining
- **Regime Robustness:** Enhanced stability during market shocks through historical experience replay
- **Market-Awareness:** Incorporation of macroeconomic indicators (MLI, PSAEWI) for context-aware optimization

2) *Model Pre-training and Architecture:* We establish our baseline model using a robust Chinese language model foundation:

As depicted in Figure 5, we utilize the ‘hfl/chinese-roberta-wwm-ext’ model from HuggingFace as our base architecture [28]. This model was selected due to its superior performance on Chinese financial texts, achieving a 92.4% accuracy on our validation set.

We begin by selecting 30,000 financial news articles, stratified by market regimes and macroeconomic events. These are used for continued pre-training of the base language model over three epochs. The loss function combines multiple domain-specific objectives:

$$\mathcal{L}_{\text{pre}} = \alpha \mathcal{L}_{\text{MLM}} + \beta \mathcal{L}_{\text{ITM}} + \gamma \mathcal{L}_{\text{Sent}} \quad (9)$$

Here, $\mathcal{L}_{\text{MLM}}$ denotes masked language modeling, $\mathcal{L}_{\text{ITM}}$ masks industry-specific terminology, and $\mathcal{L}_{\text{Sent}}$ incorporates weak sentiment signals. After pre-training, we fine-tune on 30% of labeled data for both regression and classification.

On top of the encoder, we attach two task-specific heads. The regression head is a two-layer MLP:

$$\hat{y}_{\text{reg}} = \sigma(\mathbf{W}_2 \cdot \text{ReLU}(\mathbf{W}_1 \cdot h_{[\text{CLS}]}) + b_1) + b_2 \quad (10)$$

The classification head predicts sentiment categories using:

$$\hat{y}_{\text{class}} = \text{Softmax}(\mathbf{W}_{\text{class}} \cdot \text{Tanh}(\text{Dropout}(h_{[\text{CLS}]}) \quad (11)$$

The overall loss combines the two tasks with a market-aware term:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{reg}} \cdot \text{MSE}(\hat{y}_{\text{reg}}, y_{\text{reg}}) + \lambda_{\text{class}} \cdot \text{CE}(\hat{y}_{\text{class}}, y_{\text{class}}) + \lambda_{\text{market}} \cdot \Phi(\text{MLI}_{t-\tau}, \Delta\text{PSAEWI}) \quad (12)$$

Task weights λ_k are adapted dynamically based on performance fluctuations:

$$\lambda_k = \frac{\exp(\beta \cdot \Delta s_k / \bar{s}_k)}{\sum_j \exp(\beta \cdot \Delta s_j / \bar{s}_j)} \quad (13)$$

where s_k represents task-specific metrics such as MSE or F1 score.

To stabilize training, we introduce an experience buffer $\mathcal{E}$ that stores informative samples with recency and volatility bias:

$$\mathcal{E} = \{(\mathbf{x}_i, y_i, \mathbf{m}_i) \sim P(i) \propto (1 + |r_i|)^\alpha \cdot \text{recency}(t_i)^\beta\}_{i=1}^{N_{\text{buf}}} \quad (14)$$

The buffer holds 50,000 entries, mixing 65% recent data with 35% historical analogs, with $\alpha = 1.2$ enhancing sensitivity to volatility.

Parameter updates are conditionally adjusted based on market liquidity:

$$\theta_t = \begin{cases} \theta_{t-1} - \eta_b \nabla \mathcal{L}(\mathcal{B}_t; \theta_{t-1}) & \text{if } \text{MLI} \geq \delta \\ \theta_{t-1} - \eta_b \nabla \mathcal{L}(\mathcal{B}_t \cup \mathcal{E}_{\text{sim}}; \theta_{t-1}) + \gamma \|\theta\|_1 & \text{otherwise} \end{cases} \quad (15)$$

Here, $\mathcal{B}_t$ is the current batch, and $\mathcal{E}_{\text{sim}}$ contains historical analogs. The learning rate η_b is adjusted per batch.

Lastly, the learning rate adapts to market regimes and training feedback as follows:

$$\eta_t = \eta_0 \cdot \begin{cases} \exp(0.02 \cdot \Delta \text{MLI}) & \text{if volatile} \\ \frac{2}{1 + \exp(0.1 \cdot t_{\text{stagnant}})} & \text{if plateaued} \\ 1.1 \cdot \eta_{t-1} & \nabla \mathcal{L} < \epsilon_{\text{dec}} \\ 0.6 \cdot \eta_{t-1} & \nabla \mathcal{L} > \epsilon_{\text{inc}} \end{cases} \quad (16)$$

F. Real-Time Sentiment Monitoring System

We design a real-time sentiment monitoring system capable of processing streaming financial text and adapting to dynamic market states. Incoming data are ingested through APIs (e.g., Weibo API) or automated crawlers, forming a continuous stream of social and news content. Each batch of text is processed by a fine-tuned language model to yield sentiment scores, which are stored in a MongoDB instance for retrieval and aggregation.

For user-facing insights, we implement a dashboard using the Dash framework [19], presenting sentiment dynamics at various temporal resolutions. Alerts are triggered when sentiment scores surpass critical thresholds—such as the 95th percentile—or deviate sharply according to anomaly detection models, including the isolation forest. To enhance interpretability, we integrate SHAP [30] for post-hoc attribution of

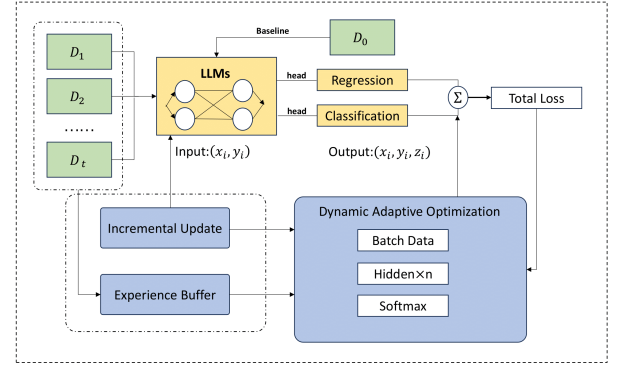


Fig. 5. Dynamic Adaptive Optimization Framework: Pre-trained RoBERTa-based model; Multi-task module with classification head and dynamic weighting

sentiment outputs, revealing which tokens or spans contributed most to model predictions. The overall streaming infrastructure is built on Apache Kafka and Apache Spark, ensuring scalability and fault tolerance.

Since sentiment dynamics vary significantly across market regimes, we tailor model regularization accordingly. Table I outlines the task-specific weights and regularization strategies used under different market conditions:

TABLE I
MARKET-STATE DEPENDENT OPTIMIZATION PARAMETERS

Regime	λ_{reg}	λ_{class}	Regularization
Bull	0.55	0.45	L1 penalty ($\gamma = 10^{-4}$)
Bear	0.40	0.60	Gradient clipping ($\tau = 2.0$)
Transition	0.65	0.35	Entropy-based penalty

The system is initialized by loading a domain-pretrained encoder ('hfl/chinese-roberta-wwm-ext') and attaching both regression and classification heads. Historical samples populate an experience buffer with 50,000 entries, structured to prioritize volatile or recent data. During operation, the system processes incoming data in 15-minute intervals:

- Step 1: Ingest batch from Kafka pipeline
- Step 2: Generate predictions and compute task loss
- Step 3: Augment batch with similar samples from buffer
- Step 4: Update model parameters with adaptive learning rate
- Step 5: Refresh buffer with new labeled samples

Model updates follow the regime-sensitive strategy defined in Eq. 15 and employ learning rates dynamically adjusted as in Eq. 16. Full validation is conducted weekly to ensure model stability. Deployed on NVIDIA A10G instances, the system achieves an average end-to-end update latency of less than 3.2 seconds per cycle.

IV. DATA

This section describes the datasets utilized.

A. Financial Text Data

Text data are sourced from Chinese financial news websites (e.g., Caixin, Sina Finance) and social media platforms (e.g., Weibo). The first part of the dataset consists of comprehensive financial news articles and key information disclosures collected from major Chinese financial media, covering topics such as macroeconomics, stocks, bonds, and mutual funds. This includes general securities industry news as well as associated commentary articles. The second part of the data is drawn from social media platforms, where financial news content is crawled from Chinese famous media accounts (e.g., official Weibo accounts of financial news outlets).

The news dataset is rich in content and captures mainstream media coverage in the Chinese financial market, with a continuous and uninterrupted sample span. Data collection employs Python's Scrapy and BeautifulSoup libraries, covering the period from January 1, 2015, to December 31, 2023, in order to encompass various market states (e.g., bull markets, bear markets). Each text entry is timestamped for alignment with market data.

Text preprocessing includes:

- Chinese word segmentation using the Jieba tool to handle the language's lack of word boundaries.
- Removal of HTML tags, special characters, and irrelevant content.
- Elimination of stop words to reduce noise.

The final dataset comprises approximately 500,000 texts, ensuring sufficient data for model training. To provide an intuitive overview of the thematic focus within the corpus. See a word cloud is presented in Figure 6.



Fig. 6. Word cloud visualization of the financial text corpus

B. Market Data

The data are structured into three key components: stock market trading indicators, individual stock data, and market-wide indicators, covering the period from January 4, 2010, to December 31, 2024. The stock market trading indicators,



Fig. 7. Closing Price(2015-2024)

sourced from the Chinese Research Data Services (CNRDS), include daily price and volume data for the Shanghai A-Share Index and the ChiNext Index, capturing the dynamics of the broader Chinese equity market and the innovation-driven sector, respectively. Individual stock data from China Stock Market & Accounting Research Database(CSMAR) database offer detailed insights into the daily trading activities of A-share listed companies. Market-wide indicators, including turnover rates, the Consumer Confidence Index, and IPO-related metrics, provide a higher-level perspective on market sentiment and activity, with correlation analyses highlighting potential multicollinearity that warrants careful consideration in regression models. Collectively, these datasets, enriched with visualizations such as time series plots, return distributions, and correlation matrices, form a robust foundation for exploring the state-dependent relationships between sentiment, market returns, and macroeconomic variables, as detailed in subsequent sections of this study.

1) *Shanghai A-Share Index and the ChiNext Index* : This section focuses on the stock market trading indicators, which encompass daily data from the Chinese equity market, spanning January 4, 2010, to December 31, 2024. The stock market trading indicators are sourced from the Chinese Research Data Services (CNRDS), a comprehensive repository of high-quality financial and economic data for China. The dataset includes daily trading records for two key indices: the Shanghai A-Share Index (ticker: 000002) and the ChiNext Index (ticker: 399006). The Shanghai A-Share Index tracks all A-share stocks listed on the Shanghai Stock Exchange, predominantly accessible to domestic investors, reflecting the performance of large and medium-sized companies. To investigate the state-dependent relationship between sentiment and returns, we first categorize market periods into Bull and Bear states based on criteria defined in III-A.

2) *Individual Stock Data*: The second part comprises individual stock data obtained from the China Stock Market & Accounting Research Database(CSMAR) database, specifically its China Stock Price dataset. This dataset includes detailed daily trading information for all A-share listed companies in China, such as stock ticker symbols, trading dates, opening and closing prices, highest and lowest prices, trading volumes, and turnover rates. These variables collectively reflect the daily

fluctuations and trading dynamics of individual stocks. The sample period spans from January 4, 2010, to December 31, 2024, thereby covering multiple market cycles and economic conditions, which enhances the robustness and generalisability of subsequent empirical analysis.

3) *Market-wide Indicators*: The third component of our market data is sourced from the China Stock Market & Accounting Research Database (CSMAR), providing a comprehensive set of market-wide indicators. These include the fund discount rate (*DCEF*, %), the previous month's trading volume (*TURN*), the number of Initial Public Offerings (*IPON*), the average first-day return of IPOs (*IPOR*), the number of new investor account openings in the previous month (*NIA*), and the Consumer Confidence Index (*CCI*). The dataset spans from January 2010 to September 2023, covering 165 monthly observations, and is temporally aligned with previously described datasets, ensuring consistency for subsequent empirical analyses. See Figure 8, the line plot illustrates the evolution of each variable, highlighting periods of volatility and stability across the sample period. For instance, *IPOR* exhibits significant peaks, such as in April 2010 (22.65%), while *CCI* shows broader cyclical movements, reflecting economic confidence shifts.

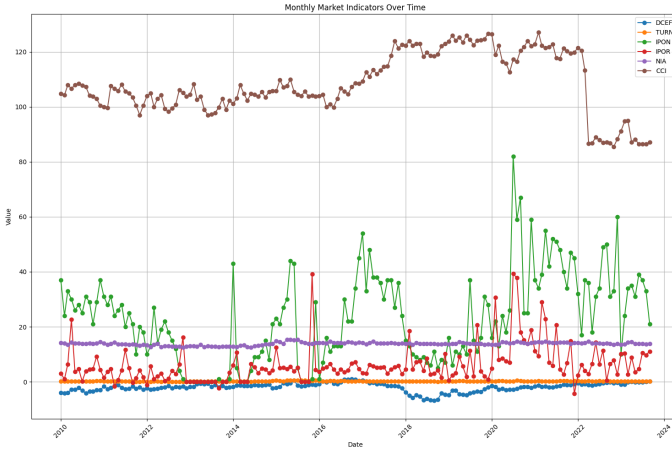


Fig. 8. Time series trends of market-wide indicators (*DCEF*, *TURN*, *IPON*, *IPOR*, *NIA*, *CCI*) from January 2010 to September 2023

C. Macroeconomic Indicators

The variables utilized in this study are sourced from CEIdata (*China Economic Information Database*), China's authoritative national economic statistics platform maintained by the State Information Center. CEIdata systematically aggregates macroeconomic indicators from over 100 official sources including the National Bureau of Statistics, People's Bank of China, and Ministry of Commerce.

We extract monthly observations spanning January 2010 to December 2025, yielding $T = 184$ time stamps that cover multiple business-cycle phases and policy regimes in China. Each series is time-stamped at the end of the reporting month to facilitate consistent alignment with our sentiment and asset-return datasets.

V. EMPIRICAL EXPERIMENT

Our next research is guided by three primary hypotheses.

H1: we suppose that LLM-based sentiment labels, along with our market indicators, can predict market returns [31] and can affect market states, with different effects in two Chinese markets.

H2: we suppose that the influence of sentiment on returns is nonlinear and varies across market states, with different effects in bull or bear markets.

H3: we suppose that integrating sentiment with market-specific indices like MLI and PSAEWI enhances predictive accuracy by capturing complementary market dynamics.

These hypotheses are motivated by prior studies highlighting the role of investor sentiment in asset pricing [1, 9] and the potential of machine learning in uncovering complex financial relationships [32].

We sampled 30,000 financial news articles processed through a fine-tuned RoBERTa-wwm-ext model to generate daily sentiment scores. Market data and Macroeconomic indicators were all cleaned and selected in the same time period of text data(2015-2023).

Two novel indices are introduced to capture specific market dynamics. The Momentum-Liquidity Index (MLI) measures short-term market momentum and liquidity conditions, reflecting immediate market responses to news and sentiment. Outlier detection is performed using the interquartile range (IQR) method, with extreme values in *daily_weighted_sentiment*, *MLI*, and *PSAEWI* replaced by their respective medians to ensure data robustness. For both indices, 12 sentiment outliers (e.g., -0.505266 on day 11, 0.540570 on day 457), 148 MLI outliers (e.g., 5.038305 on day 74), and one PSAEWI outlier (1.516909 on day 918) are identified and adjusted. The market state (*market_state*) is defined using a 250-day moving average, with values of 1 for bull markets (closing price above the moving average) and 0 for bear markets.

A. H1 & H3: Model Performance

Random Forest and XGBoost models are employed to capture complex nonlinear relationships and interactions, with 100 trees and early stopping (10 rounds) for XGBoost. Feature importance is extracted to identify key predictors. And we also use some econometric model:

- **Interaction Regression**: Tests state-dependent effects between sentiment and market states:

$$R_t = \beta_0 + \beta_1 S_t + \beta_5 (S_t \cdot MS_t) + \epsilon_t \quad (17)$$

- **Quantile Regression**: Examines distributional effects of sentiment across return quantiles:

$$Q_\tau(R_t | S_t, MLI_t, PSAEWI_t, MS_t) = \beta_0(\tau) + \beta_1(\tau) S_t + \dots \quad (18)$$

- **XGBoost & Random Forest**: Machine learning model that captures nonlinear relationships and high-order interactions among predictors.

- **Granger Causality Test:** Evaluates whether *daily_weighted_sentiment*, *MLI*, or *PSAEWI* Granger-cause returns using a vector autoregression (VAR) model with a maximum lag of 5:

$$\mathbf{Y}_t = \mathbf{A}_1 \mathbf{Y}_{t-1} + \mathbf{A}_2 \mathbf{Y}_{t-2} + \cdots + \mathbf{A}_5 \mathbf{Y}_{t-5} + \epsilon_t, \quad (19)$$

where $\mathbf{Y}_t = [R_t, S_t, MLI_t, PSAEWI_t]$.

Table II provides robust evidence supporting **H1**, confirming that LLM-based sentiment indicators, together with market-level variables, significantly predict future returns in Chinese equity markets. The notably higher R^2 values of machine learning models (above 0.84) demonstrate the presence of non-linear and complex relationships that traditional regressions fail to capture. This highlights the ability of sentiment features derived from large language models to capture latent investor expectations and behavioral biases that are otherwise omitted.

Moreover, the significance of PSAEWI and MLI in both Granger causality and VAR models at lag 5 confirms their delayed predictive utility. Also we take advantage of them as complementary so that our result (R^2) is nearly 4 times as high as other researcher’s findings imply [31]. In line with **H3**, these results demonstrate that combining sentiment with structural indicators like MLI and PSAEWI enhances forecast accuracy by capturing complementary sources of market information—emotional reactions and institutional signals—thereby improving the robustness of return prediction frameworks.

B. H2: Asymmetric Sentiment Effects

Figure ?? demonstrates the bull market amplification effect: sentiment coefficients increase by 106% (ChiNext) and 83% (Shanghai) in bullish conditions. This confirms the nonlinear state-dependent relationship hypothesized in H2.

Bull and bear market analyses highlight sentiment’s role in bull markets. For Shanghai, sentiment is significant (coefficient=0.7523, $p=0.001$, $R^2=0.011$), while for ChiNext ($n=736$), it is stronger (coefficient=1.4720, $p=0.003$, $R^2=0.013$), as shown in Figure ?. Bear markets show no significant predictors ($R^2 \leq 0.002$).

C. Further discussion: Prediction of macro-performance indicator

Based on the theoretical framework established by Benhabib et al.[33], investor sentiment not only influences asset prices and returns but also impacts macroeconomic fundamentals. Their model demonstrates that optimistic financial market sentiments can lead to increased capital prices, signaling strong economic fundamentals, which in turn stimulate real output and employment. This mechanism illustrates how sentiment-driven fluctuations can result in self-fulfilling business cycles. In this section, we empirically test whether sentiment variables derived from large language models (LLMs) possess predictive capabilities concerning macroeconomic indicators, aligning with the sentiment theory characteristics outlined above.

We use HAC-OLS regression to evaluate the sentiment index’s ability to predict key Chinese macroeconomic indicators:

$$Y_t = \alpha + \beta_1 \text{Sentiment}_{t-1}^{\text{std}} + \beta_2 \text{MLI}_{t-1} + \beta_3 \text{PSAEWI}_{t-1} + \phi Y_{t-1} + \epsilon_t, \quad (20)$$

where the disturbance ϵ_t may feature conditional heteroskedasticity and mild serial correlation; inference therefore relies on Newey–West standard errors with four lags. HAC-OLS is well suited to low-frequency macro data because it yields consistent t -statistics under a broad array of covariance structures [13, 27].

We lists coefficients, t -statistics and fit metrics for the three macro variables whose sentiment slopes are significant at the five-percent level in Table III. A clear economic narrative emerges. The negative link between lagged sentiment and subsequent growth in on-balance-sheet social financing indicates that episodes of optimism are followed by a partial retreat of bank-intermediated credit, echoing the substitution channel proposed by Zhang and Wang [6]. Conversely, the positive association with *transport_investment* suggests that buoyant sentiment reduces borrowing costs for local-government financing vehicles, triggering infrastructure spending—a fiscal front-loading pattern documented by Liu et al. [16]. Finally, the negative coefficient on the Purchasing Managers’ Index conforms with the sentiment-front-loading story of Antweiler and Frank [17]: business managers place orders early when optimism prevails, leaving fewer orders in the subsequent month and mechanically depressing the PMI.

In Figure 9, sentiment surges in 2018 and 2022 precede downturns in CPI by roughly two quarters. Figure 10 demonstrates an inverse co-movement between sentiment and closed-end-fund discounts, replicating the U.S. findings of Baker and Wurgler [13]. Finally, Figure 11 overlays sentiment on the money supply: the monotonic rise in $M2$ reflects structural deepening, but cyclical humps align with sharp sentiment swings, hinting that monetary authorities inject liquidity when public mood deteriorates. So, the LLM sentiment index

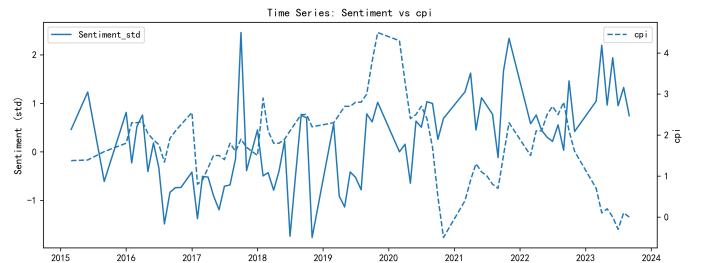


Fig. 9. Dual-axis time series of the sentiment index and CPI

offers a behavioural early-warning indicator that augments backward-looking macro statistics. A sharp move in the index—especially when accompanied by a liquidity pulse in MLI—may flag imminent swings in manufacturing output or infrastructure spending, guiding counter-cyclical action. For investors, the documented relations justify blending sentiment with liquidity and valuation screens in tactical macro strate-

TABLE II
MODEL PERFORMANCE COMPARISON (R^2 , MSE, SIGNIFICANCE)

Model	Market	R^2	MSE	Significant Predictors
Interaction Regression	ChiNext	—	—	Sentiment $\times$ Market State ($p=0.026$)
Interaction Regression	Shanghai	—	—	Not significant ($p=0.203$)
Quantile Regression ($q=0.9$)	Shanghai	—	—	Sentiment ($p=0.005$), MLI ($p=0.035$)
Quantile Regression ($q=0.487/0.5$)	ChiNext	—	—	MLI ($p=0.487$), MS ($p_i0.001$)
Random Forest	Shanghai	0.8323	0.2959	—
XGBoost	Shanghai	0.8405	0.2815	—
Random Forest	ChiNext	0.8421	0.5907	—
XGBoost	ChiNext	0.8500	0.5612	—
Granger Causality (Lag 5)	Shanghai	—	—	PSAEWI ($p=0.0057$), MLI ($p=0.0401$)
Granger Causality (Lag 5)	ChiNext	—	—	PSAEWI ($p=0.0108$)
VAR (Lag 5)	Shanghai	—	—	PSAEWI ($p_i0.001$), MLI ($p=0.027$)
VAR (Lag 5)	ChiNext	—	—	PSAEWI ($p=0.002$)

TABLE III
HAC-OLS ESTIMATES FOR THE THREE MACRO VARIABLES WITH THE MOST SIGNIFICANT SENTIMENT COEFFICIENTS

	Financing Growth	Transport Investment	PMI
β_1 Sentiment $_{t-1}$	-0.152^{**} (0.069)	15.18^{**} (7.05)	-0.227^{**} (0.101)
β_2 MLI $_{t-1}$	0.068	1.70	-0.090^{**}
β_3 PSAEWI $_{t-1}$	-0.091	12.88^{**}	-0.101
ϕ Y_{t-1}	0.912^{***}	0.477^{***}	0.597^{***}
Adj. R^2	0.914	0.350	0.440
Observations	78	78	78

Note: Robust Newey–West standard errors in parentheses. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

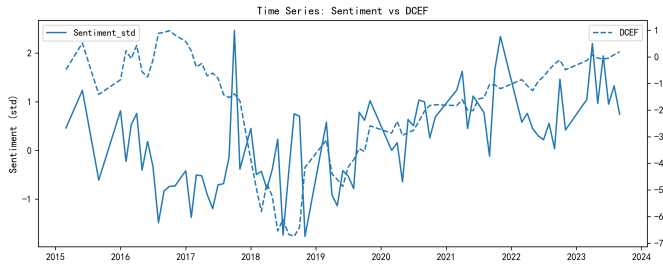


Fig. 10. Dual-axis time series of the sentiment index and closed-end-fund discount (DCEF)

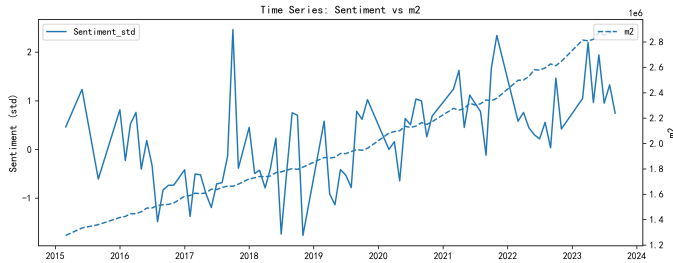


Fig. 11. Dual-axis time series of the sentiment index and broad money stock ($M2$)

gies, echoing FinBERT applications in portfolio design [4]. More broadly, the study corroborates the thesis that when fine-tuned on domain corpora, LLMs yield textual signals that are both statistically robust and economically meaningful. As real-time data streams proliferate, embedding such behavioural metrics into standard macro-forecasting toolkits represents a fruitful frontier for research and practice alike.

VI. CONCLUSION

In this paper, we establishes that LLM-derived sentiment significantly enhances market return prediction and macroeconomic forecasting in China's financial markets. Empirical results reveal three key findings: First, sentiment exhibits asymmetric effects across market states, with significantly stronger predictive power during bull markets (Shanghai: $\beta = 0.7523$, $p = 0.001$; ChiNext: $\beta = 1.4720$, $p = 0.003$) compared to bear periods. Second, integrating sentiment with our novel Momentum-Liquidity Index (MLI) and Price Stability Adjusted Equal-Weight Index (PSAEWI) substantially improves forecasting accuracy, achieving $R^2 = 0.85$ in XGBoost models. Third, sentiment serves as a leading indicator for macroeconomic dynamics, significantly predicting financing growth ($\beta = -0.152$), transport investment ($\beta = 15.18$), and PMI ($\beta = -0.227$), with observed 2-quarter leads on CPI fluctuations.

For policymakers, these findings suggest concrete applications: Regulators should integrate sentiment indices with market liquidity signals into real-time monitoring frameworks like our Apache Kafka/Spark prototype for early detection

of systemic risks, particularly when sentiment-M2 deviations signal liquidity mismatches. Central banks could enhance monetary policy responsiveness by incorporating sentiment-implied behavioral shifts, especially to anticipate fiscal front-loading effects in infrastructure spending. Financial institutions should adopt the Dynamic Adaptive Optimization (DAO) module's regime-sensitive weighting for microcredit risk assessment, reducing non-performing loans during market transitions through continuous parameter recalibration.

Future research should prioritize three directions: 1) Multi-source sentiment integration combining news with social media (Weibo/X) and satellite imagery via federated learning; 2) Cross-asset validation extending our dual-model framework to bond markets and cryptocurrencies where retail sentiment dominates; and 3) Real-time system optimization through edge-computing implementations of our monitoring prototype for latency-sensitive applications. Additionally, sectoral heterogeneity in sentiment transmission—particularly between strategic sectors like green energy and property—warrants deeper investigation to refine policy targeting.

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APPENDIX

The supplementary figures and some database or platforms where we got data.

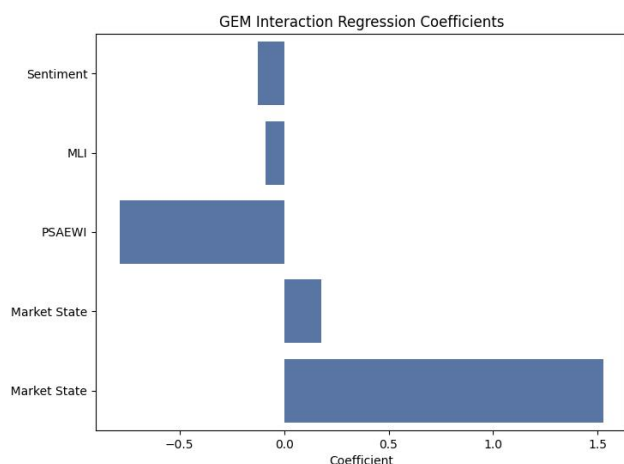


Fig. 12. Interaction Regression Coefficients for ChiNext

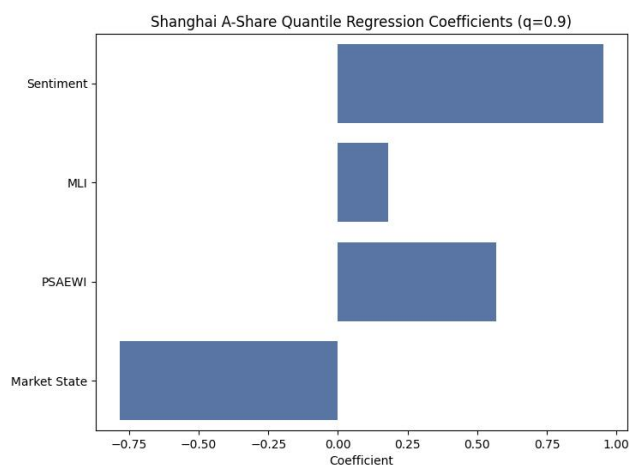


Fig. 13. Quantile Regression Coefficients for Shanghai A-Share (q=0.9)

- CEIdata (*China Economic Information Database*)
- Financial news portals such as Caixin and Sina Finance
- Social media platforms, e.g., Weibo

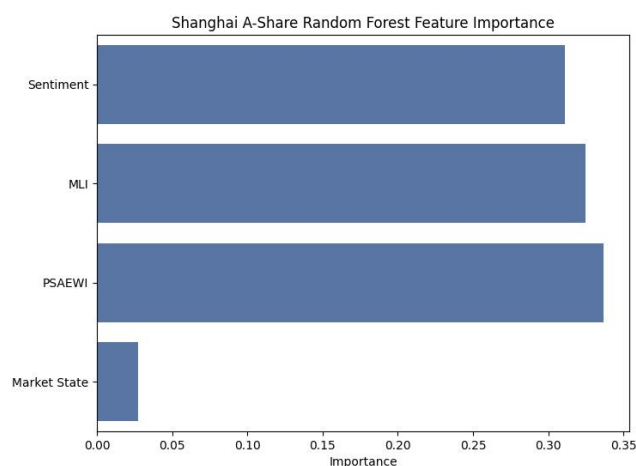


Fig. 14. Random Forest Feature Importance for Shanghai A-Share

- Chinese Research Data Services (CNRDS)
- China Stock Market & Accounting Research Database (CSMAR)
- CNEconDic: an openly available Chinese financial lexicon containing 23.517 curated terms covering macroeconomics, equity, fixed income, derivatives, and regulatory jargon.